TS5000 Robot Controller

TS5000-MS

Interface Manual

(Compatible Robots: THE600, THE800, THE1000)

Rev.8

March 17, 2025

IMPORTANT

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

This instruction manual applies to the controller TS5000-MS.

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Preface

This manual describes the types of external connection cables, their functions, and how to use them when connecting the TS5000-MS robot controller to an external device. This manual is intended for system designers and manufacturing engineers. The TS5000-MS can control digital input/output signals using SCOL2 language (robot language for creating programs). Also, this interface includes system input signals that enable usage of a controller from an external device, system output signals for notifying about the controller state (sequence processing function), and a communication function using TCP/IP protocol for enabling the user to easily build an FA system.
Before using this manual, be sure to read “STEA1179_THE Series Safety Manual” and “STEA1246_THE800/ THE1000 Safety Manual” to ensure that you fully understand the safety measures.

This manual consists of the following sections.

Chapter 1: Overview
This presents an overview of the controller interface.

Chapters 2 to 12: Details
This provides a detailed description of the components of the controller interface (power cables, robot control cables, input/output external power supply and safety input/output cables, input signal cables, output signal cables, TP cables, SD card slot, Ethernet connector, extension I/O slot).

Chapter 13: I/O Cables
This describes the INPUT cable and OUTPUT cable connected to the controller.

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Usage Scope

This system was designed for supporting automation of a wide range of operations, including transportation operations such as workpiece movement and alignment and assembly operations such as part insertion and screw tightening.
We ask that customers use this system for the purpose of supporting automation of these operations and incorporating into automation equipment.

This system was not designed for use in operations where contact is made with people (operations in medical treatment, nursing care, etc.), for incorporating into equipment used for sustaining human life (life support devices, infant incubators, etc.), for incorporating into equipment used for transporting people (automobile and other transportation equipment, airplanes and other aerospace equipment, roller coasters and other amusement park equipment), for equipment used for protecting human life or the human body (safety devices, etc.), or similar equipment.
This system was also not designed for incorporating into equipment for the purpose of controlling high amounts of heat (nuclear control equipment, combustion furnace control equipment, etc.).
In cases where a fault or malfunction of this product could directly threaten human life or cause an injury, its use in operations or incorporating into equipment is prohibited. If this product is used, the customer assumes all responsibility for ensuring safe operation.

Operating Procedures

This section describes the conditions of use for robots that can be connected to the TS5000-MS. Using the product in a way that is different from the conditions may cause unexpected risks. Be sure to follow the required conditions.

1. Main Specifications of Product Refer to the robot manual for your robot.
2. Usage Restrictions The "Restricted Specifications Sheet", which shows the usage restrictions, are contained in Tables 1 to 6.

Table 1. Restricted Specifications Sheet

1. Purpose and application for using product (usage restrictions)	
1) Intended use	To make the robot perform operations for specific purposes (item transport, assembly, inspection, machining, etc.) (1) Transportation [1] Obtain a transportation route to the unpacking site. [2] Use a forklift or other equipment to lift the packing box. [3] Move the packing box to the unpacking site. (2) Unpacking [1] Obtain the space required for unpacking. [2] Cut the packing straps and unpack. [3] Cut the packing straps securing the robot. [4] Peel off the tape from the controller packing box, and unpack. [5] Check that the robot and controller are not damaged. [6] Check the accessories and their quantities. - When transferring the robot to the installation site [7] Dispose of, or store and manage the packing materials. - When storing the robot [7] Secure the robot in place in the same way as the state at delivery, and repack. [8] Move the packing box to the storage site. (3) Transfer - Robot [1] Check the orientation of the robot during transfer. [2] Check that the robot fixing jigs are mounted. [3] If using a crane, mount the supplied eyebolts. [4] Check the tightness of the robot fixing jig fixing bolts. [5] Use a crane or cart to transfer the robot to the installation site. - Controller [1] Use a cart to transfer the controller to the installation site.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(4) Installation [1] Use bolts to secure the robot to the frame and other parts. [2] Fabricate any needed cables. [3] Remove the robot fixing jigs and eyebolts. [4] Install the controller and connect it to the robot. [5] Push the robot by hand to move to a location where it is easy to perform work. [6] Mount the hand effector. [7] Connect the required external devices. (5) Teaching [1] Move the robot to the desired position or orientation by guiding it with a teach pendant or pushing it by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation [1] Start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [1] Remove the hand effector. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Obtain the space needed for packing. [6] Use a crane to transfer the robot, and mount the supplied eyebolts. [7] Transfer the robot and controller to the packing site. [8] Pack the robot and controller so that they are in the original delivery state. [9] Obtain a transportation route to the transportation equipment (truck, etc.). [10] Use a forklift to lift the packing box. [11] Load the packing box onto the transportation equipment.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance Work performed when the primary power is off: - Robot [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check for any wear on the cable and air tube. [7] Check that there is no misalignment between the motion of each joint axis and the origin position. - Controller [1] Replace the memory backup batteries in the controller. Work performed when the primary power is on: [1] Check the grease state and apply grease (when the servo power is off). [2] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [3] Check the teaching point for any misalignment (when the servo power is on). [4] Replace the backup batteries for the robot position detector (when the servo power is off). (9) Disposal [1] Remove the hand effector. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Use a crane to transfer the robot, and mount the supplied eyebolts. [6] Transfer the robot and controller to the disposal site. [7] Dispose of the robot using a method compliant with your country’s laws and regulations.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot - Modification of robot, controller, and optional parts, or supplying power to robot, controller, and optional parts while their covers are removed - Operation in other installation state than the specified (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that operations would have stopped although the stop is just for waiting for an operation signal during operation - Usage under conditions where the specification values are exceeded or usage under improper settings - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by inadvertently pressing of a button - Operating without checking operation in MANUAL mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Malfunction due to failure or other cause

Table 2. Restricted Specifications Sheet

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for five (5) minutes - Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).
2) Electrical restrictions	

Table 3. Restricted Specifications Sheet

3. Movable range of product (restriction on space)		
1) Work envelope	1st axis 2nd axis 3rd axis 4th axis Additional axis - -	Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with individual specifications. - -
2) Interface	Transportation: Unpacking: Transfer: Installation: Teaching: Operation: 	The packing box shall be able to be transferred by a transportation worker. The packing box shall be able to be carried in. Space of at least 2 m^3 shall be secured. The robot shall be able to be hoisted by a crane. The robot shall be able to be lifted by two or more people and loaded onto a cart or similar device. There shall be no obstacles on the route from the unpacking site to the installation site. The product shall be installed at a location suitable for the outline and work envelope of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cable shall be installed at a location where it will not be stepped on by operators. Teaching work shall be able to be performed from outside the hazard zone of the robot. Space shall be provided to prevent a person from getting caught between the robot and peripheral equipment in case the person enters the hazard zone. A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location.

Table 3. Restricted Specifications Sheet (continued)

3. Movable range of product (restriction on space)		
2) Interface (continued)	Repacking:	Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. The repacking work shall be able to be performed. A forklift or other equipment shall be able to be used to lift the packing box.
	Maintenance:	Space required for maintenance shall be obtained. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. Space shall be provided to prevent a person from getting caught between the robot and peripheral equipment in the hazard zone.
	Disposal:	Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the zone during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be emitted.	

Table 4. Restricted Specifications Sheet

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, risk assessment is carried out for the life cycle of “transfer and installation” (transportation, unpacking, transfer, installation, and repacking), “use” (teaching, operation, and maintenance), and “disposal.”

Table 5. Restricted Specifications Sheet

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
1) Transportation worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for forklifts and other transfer equipment work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
2) Unpacking worker	Knowledge	Contents of the instruction manual
	Requirement	-
3) Transfer worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
4) Installation worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
5) Teaching worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	Person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
6) Operator	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	-
7) Repacking worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer

Table 5. Restricted Specifications Sheet (continued)

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
8) Maintenance worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations Storage procedures for target robot
	Requirement	Person who has received training for robot testing work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
9) Disposal worker	Knowledge	Contents of instruction manual, operating procedures for target robot, and waste disposal methods
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
10) People passing nearby Other workers	- Measures shall be taken to prevent entry into the fence, enclosure, or other protection surrounding the hazard zone of the robot. - Measures shall be taken to make the robot stop operation when someone approaches the hazard zone. - If another worker enters the hazard zone of the robot to perform work, robot operation shall be stopped, the drive power turned off, and a teaching worker or maintenance worker shall supervise the situation to ensure safety.	

Table 6. Restricted Specifications Sheet

6. Other

Warranty

This product is provided by SHIBAURA MACHINE after strict testing and inspection and verification that it satisfies our performance standards.
SHIBAURA provides the following warranty in the event that a fault occurs.

1. Warranty Period

The warranty period of this product is one of the following periods, whichever is shortest.

(1) Period of 18 months from SHIBAURA MACHINE factory shipping

(2) Period of 12 months from customer installation

2. Warranty Terms and Conditions

(1) The warranty applies to this product. The warranty scope is the specifications and functions described in the product specifications, catalog, instruction manual, and other documents. SHIBAURA MACHINE will not be liable for any secondary or incidental damages arising as a result of faults in this product.

(2) Within the warranty period of this product, SHIBAURA MACHINE will repair free of charge only those faults that arise under operation and use in accordance with the instruction manual provided with this product.
A fee will be charged for all repairs of faults that occur after the warranty period has expired.

3. Disclaimers

The following cases are outside the scope of the warranty.

(1) Faults and damage due to usage that did not follow the instruction manual or due to negligent usage.

(2) Problems due to deterioration over time or usage wear (natural fading of the paint coating, deterioration of consumable parts [1)]

(3) Problems due to sensory experiences (such as occurrence of noise that does not affect functionality)

(4) When the product was modified or disassembled without the approval of SHIBAURA MACHINE

(5) Faults and damage due to inadequate inspection and maintenance or improper repairs

(6) Faults and damage due to natural disasters, fires, and other external factors

(7) Internal data such as programs and points created or modified by the customer

(8) When this product was purchased in Japan and taken outside the country

1) Consumable parts refer to the maintenance replacement parts shown in chapter 7 of this instruction manual.

4. Notes

(1) SHIBAURA MACHINE cannot guarantee the basic performance of this product when it is used in an environment outside the specifications.

(2) SHIBAURA MACHINE will not be liable for any injuries, damage, faults, or other problems that arise by failing to follow the warning and caution information provided in this instruction manual.

(3) Be aware that the warning, caution, and other safety information in this instruction manual apply to the safety risks foreseen by SHIBAURA MACHINE, and they do not cover every possible scenario.

Main standards, laws, and regulations pertaining to the safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" describes the risk reduction measures for machines and equipment.
Step 1) Inherently safe design measures: Limitations on force, speed, energy, etc.
Step 2) Safeguarding and/or complementary protective measures: Installation of safety fence, etc.
Added protective measures: Installation of emergency stop devices, etc.
Step 3) Information for use: Warning signage, alarms, instruction manuals, etc.

Based on this, the guidelines known as "Guide 51" were defined by classifying the various standards for the ISO/IEC international standards in a hierarchical structure.
The safety standards for industrial robots are standardized as type C in the individual safety standards for machinery.

- ISO 10218 and ISO 10218-1

Be sure to follow the standards, laws, regulations established in the country where the industrial robot is used.

Cautions on Safety

The instruction manual for the controller contains important safety information for ensuring the safe and correct operation of this device and for preventing hazards to the operators and others and property damage.

To ensure that the work procedures from installation to operation of the controller are performed safely, be sure to carefully read the separate instruction manual “STEA1179_THE Series Safety Manual” and “STEA1246_THE800/THE1000 Safety Manual” before starting the actual work.

Make sure that you well understand the following details (indications and symbols) before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
⚠ DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
⚠ WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
⚠ CAUTION	Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury[1)].
NOTICE	Indicates a situation which could result in physical damage[2)] only.

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

⚠ DANGER
● To ensure safe operation of the controller from transport to disposal, be sure to carefully read “STEA1179_THE Series Safety Manual”, and “STEA1246_THE800 and THE1000 Safety Manuals”. Operating the controller differently from the instructions described in the manual may cause harm to human body.

Service Network

Please refer to our distributor's handouts.

Overview

This presents an overview of the controller interface.

1. Overview of Interface

Controller maintenance and inspection must be performed by qualified personnel.
Also, these processes must be compliant with the laws, regulations, and ordinances of the respective country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

The TS5000-MS robot controller is connected with the robot and external equipment, using connectors and terminal blocks provided on the front and rear panels of the controller. For details on the connectors, refer to “STEA1161_TS5000-MS Controller Manual”.

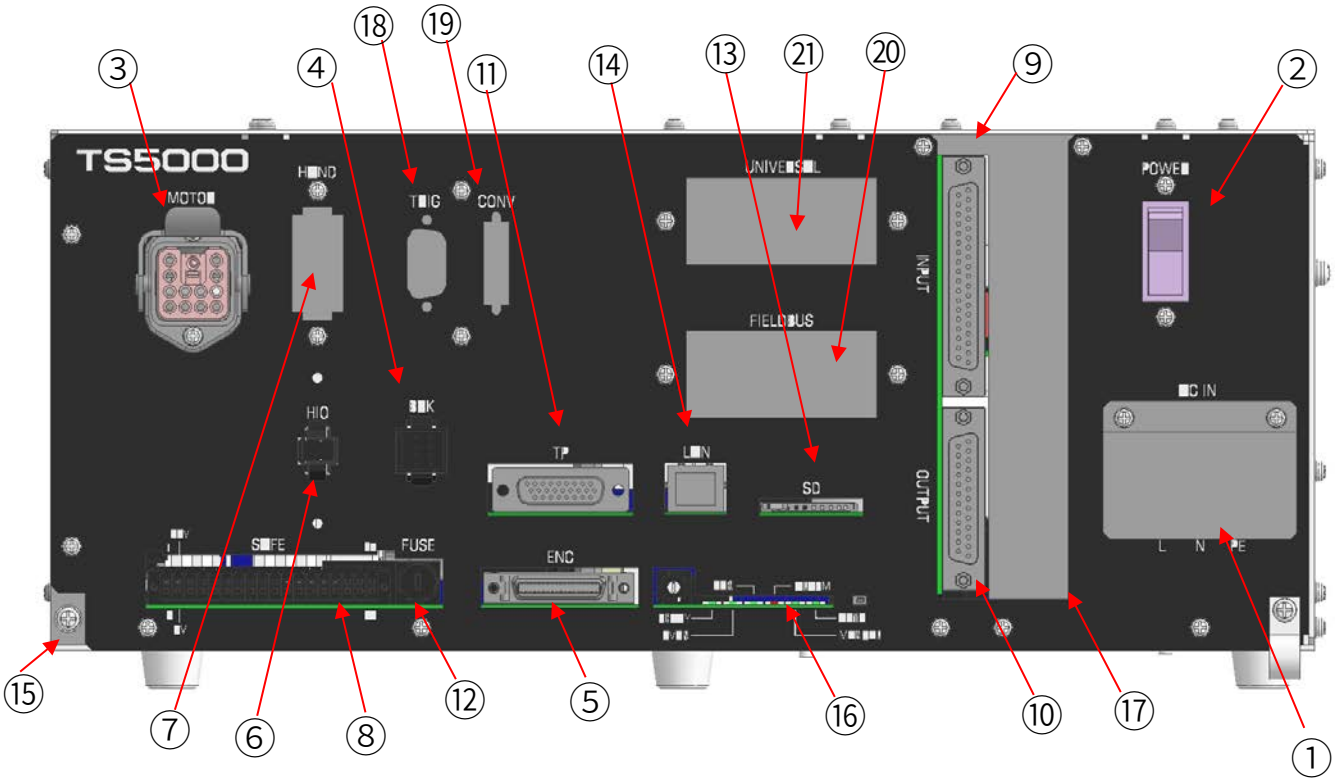


Fig. 1.1 Layout and name of connectors (TS5000-MS front view)

1.1 Power Supply Terminal "AC IN" Fig.1.1-①

The power cable is an interface that supplies the main power (single phase 200 VAC 50/60 Hz) to the TS5000-MS Robot Controller. The connector "ACIN" is used. For details, refer to 2. Power Cable.

1.2 Power Supply Circuit Breaker "POWER" Fig.1.1-②

This is the single-phase 200 VAC power supply circuit breaker for the robot controller.

1.3 Robot Control Cables

In the standard configuration, the cables between the robot and controller are the motor drive cable, brake cable, encoder cable, and hand I/O (communication) cable (total of four cables).

1.3.1 Motor Drive Cable "MOTOR" Fig.1.1-③

This connects the TS5000-MS Robot Controller to the robot body and supplies three-phase AC power to the robot axis motors. This cable uses the MOTOR connector. For details, refer to 3.1 Motor Drive Cable.

1.3.2 Brake Cable "BRK" Fig.1.1-④

This connects the robot controller to the robot body. This cable controls the parking brake signal that is sent to the robot axis motors for locking the axes. This cable uses the BRK connector. For details, refer to 3.2 Brake Cable.

1.3.3 Encoder Cable "ENC" Fig.1.1-⑤

This connects the TS5000-MS Robot Controller to the robot body and controls the encoder signals for detecting the rotation angle of the robot body axes. This cable uses the ENC connector. For details, refer to 3.3. Encoder Cable.

1.3.4 Hand I/O Communication Cable "HIO" Fig.1.1-⑥

The TS5000-MS robot controller is connected to the robot body to control I/O wired to the hand at the tip of the robot. Standard input/output of 8 points/8 points are available, and the controller and the robot are connected using an HIO connector. The connection between the robot and the customer is wired via the "HAND" connector on the second arm of the robot. For details, refer to 3.4. Hand I/O Cable. This slot is used when THE600, THE800, or THE1000 is connected. When this connector is used, the HAND slot is not used.

1.3.5 Hand I/O Cable "HAND" Fig.1.1-⑦

* Supported as needed

This can be used by inserting the HAND connector into the HAND slot. This connects the TS5000-MS Robot Controller to the robot body. This connector is used when connecting the compact SCARA robot (support planned if needed). The hand I/O communication cable controls the hand on the robot tip, and in its standard configuration, is capable of using 8 inputs and 8 outputs. This cable uses the HAND connector.

When this connector is used, the HIO slot is not used.

1.4 Input/Output External Power Supply and Safety Input/Output Cable "SAFE" Fig.1.1-⑧

The SAFE connector is used to connect emergency stop signal input and safety door signal input to the robot controller from an external device and to connect an external power supply cable for I/O signals. It has redundant c-contact outputs which separately show the controller emergency stop status and safety door status for enabling easy connection to upper-level sequencers.

Also, this connector is used to supply 24 VDC for I/O signals, and so be aware that if it is not connected, the 008-278 Ext DC24V Regulator error will occur. For details, refer to 4. Input/Output External Power Cable and Safety Input/Output Cable.

1.5 Input Signal Cable "INPUT" Fig.1.1-⑨

The input signal cable is an interface that recognizes signals input to the robot controller from an external device.

The input signals include 8 general-purpose input signals that can be used by the user for programming using SCOL language or the sequence function in addition to 13 system input signals that allow use of the controller from an external device. A servo ON system input signal and a servo OFF system input signal are also included in this cable. For details, refer to 5. Input Signal Cable.

1.6 Output Signal Cable "OUTPUT" Fig.1.1-⑩

The output signal cable is an interface that performs output of signals from the robot controller to an external device.

The system output signals include 8 general-purpose output signals that can be used by the user for programming using SCOL language or the sequence function in addition to 9 system output signals that output the controller status to an external device. For details, refer to 6. Output Signal Cable.

1.7 Teach Pendant Cable "TP" Fig.1.1-⑪

This is a serial port connector for the teach pendant (TP), and it connects the robot controller and TP. The TP is an option. The TP can be used to create programs, perform manual guidance of the robot, and other operations. The connector "TP" is used.

The connectible TPs are the TP5000 and TP1000 (special conversion cable is required). For details, refer to 7. Teach Pendant Cable

1.8 Fuse Holder "FUSE" Fig.1.1-⑫

This is a fuse for overcurrent detection (3 A) of the digital signal input/output to the robot controller. It is used to protect patterns. "FUSE" is located at the bottom of the front panel. For details on fuse replacement, refer to “STEA1161 TS5000-MS Controller Manual”.

1.9 SD Card Slot "SD" Fig.1.1-⑬

An SD card can be used to transfer and save parameters and to back up programs and other data. For details, refer to 8. SD Card Slot.

1.10 Ethernet Connector "LAN" Fig.1.1-⑭

This connector supports connection of a robot controller using 1000BASE-T. The robot controller uses an automatic recognition function to enable usage of either a straight-through cable or crossover cable. Also, TCP/IP protocol can be used to enable communication with computers or visual devices on the Ethernet within the same local area, and the program assistance software TSAssist can be used to perform program creation and editing, assist in debug startup, and perform maintenance. For details, refer to 9. Input Signal Cable. Ethernet Connector.

1.11 Ground Terminal Fig.1.1-⑮

This is a PE ground terminal of the controller for preventing electric shock. Connect the PE using a 2.0 mm^2 cable whose length is 1 m or shorter with a green grounding wire or a green and yellow grounding wire. Do not attach any switch or overcurrent protection device to the PE cable.

1.12 LED Display Unit Fig.1.1-⑯

LEDs are provided for indicating the controller status. A silkscreen printing is provided on the controller that reads READY SVON RUN ALARM MANUAL AUTO.

“READY” LED: Turns on when the controller was started normally.
“SVON” LED: Turns on when the servo is ON.
“RUN” LED: Turns on when a program is being executed.
“ALARM” LED: Turns on when an alarm has occurred.
“MANUAL” LED: Turns on when MANUAL mode is selected.
“AUTO” LED: Turns on when AUTO mode is selected.

1.13 Extension I/O Slot (Option) Fig.1.1-⑰

The extension I/O slot is used for inserting I/O printed boards. The I/O printed board is capable of adding 21 general-purpose inputs and 17 outputs. For details, refer to 10. Expansion I/O Slot (Option).

1.13.1 Extension Input Signal Cable “EXP-IN”

The extension input signal cable is an interface that recognizes external signals input to the robot controller from an external device. A total of 21 extension input signals can be set by the user using SCOL2 language or a ladder program.

1.13.2 Extension Output Signal Cable “EXP-OUT”

The extension output signal cable is an interface that performs output of external signals from the robot controller to an external device. A total of 17 extension output signals can be set by the user using SCOL2 language or a ladder program.

1.14 TRIG Slot "TRIG" (Option) Fig.1.1-⑱

The TRIG slot is for extension of the conveyor synchronization trigger input. The trigger input signal cable provides 4 inputs for the trigger input function. It is used for the data latch function, conveyor synchronization function, and other functions. For details, refer to 11 TRIG Slot (Option).

1.15 CONV Slot "CONV" (Option) Fig.1.1-⑲

The CONV slot is for extension of the conveyor synchronization encoder input. The conveyor synchronization signal cable is for the conveyor synchronization function. For details, refer to 12 CONV Slot (Option).

1.16 FIELDBUS Slot "FIELDBUS" (Option) Fig.1.1-⑳

The FIELDBUS slot is provided for FIELDBUS connection. Fieldbus slave modules can be selected from one of six types: CC-Link, EtherNet/IP, EtherCAT, PROFINET, DeviceNet, and PROFIBUS. For details, refer to “STEA1190_Fieldbus Slave Manual”.

1.17 UNIVERSAL Slot "UNIVERSAL" (Option) Fig.1.1-㉑

* Unused

Details

This presents the details of each interface of the controller.

2. Power Cable

The power cables must be connected by qualified personnel. Also, these processes must be compliant with the laws, regulations, and ordinances of the respective country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

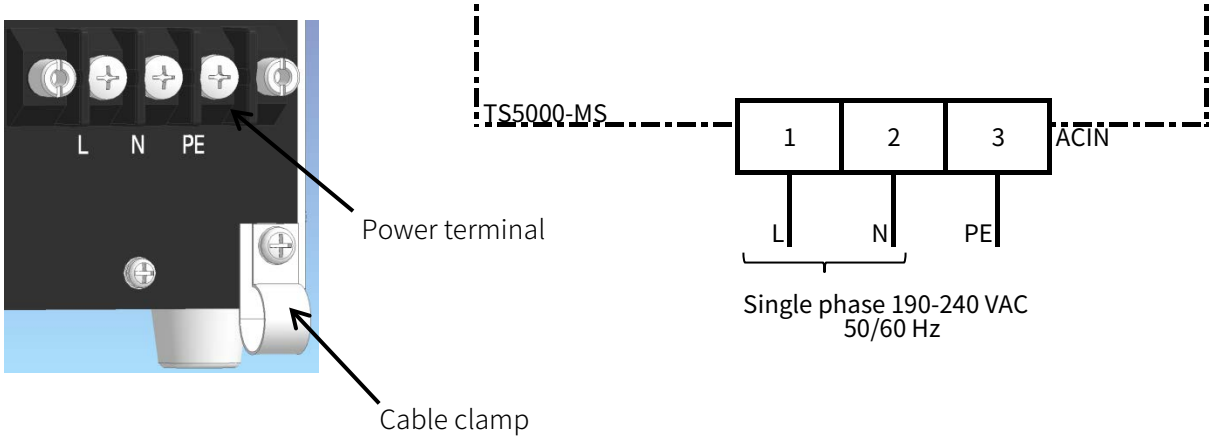
⚠DANGER

- Never use unauthorized cables. Manufacturing and use of unauthorized cables can result in a fire.
- Do not insert connectors or connect to terminals that are not specified. Unspecified wiring causes fires. Also, electric shock may occur due to leakage of electricity, etc.
- Do not insert conductive objects or your fingers into the power supply terminal parts. An electric shock could occur.
- Make sure to connect a ground wire. If the ground wire is not connected, there is a risk of electric shock due to leakage of electricity.
- After wiring, use a voltage tester to check the voltage and short-circuit to avoid potential problems. If the wiring is incorrect or if the controller is malfunctioning, turning on the power may cause a fire. Also, electric shock may occur due to leakage of electricity, etc.
- Mating should be done securely. If the mating is loose, static electricity may enter and cause the robot to malfunction and collide.

2.1 Connecting the Power Cable

Use round terminals for the power supply terminals when connecting the power cable. At the M4 terminal, connect the wiring in the order of L, N, PE from the left. After wiring, secure the power supply line with a cable fixing clamp. The cable is not included in the accessories. The customer must fabricate the cable.

Plug connector model: OGM-300NE-3P*D Osada Co., Ltd.
Recommended wire: VCTF cable 2.0 mm^2 (AWG#14) Voltage-resistant 300 V 23A or less, 60°C or below



Pin No.	Signal name	Signal description
1	L	200 VAC, L phase
2	N	200 VAC, N phase
3	PE	200 VAC, FG

Table 2.1 Specification of power cable

Power supply	Single-phase 190-240 VAC 50/60 Hz ± 1 Hz
Power capacity	Varies depending on the robot to be connected.
Instantaneous power failure	40 msec or less
Ground	Class D dedicated grounding
Size of connecting screw	Use M4 round terminals for connection.

A ground-fault circuit interrupter must always be installed by the customer.

Specifications: Single phase 30A, ground-fault circuit interrupter of sensitivity current 100mA

Recommended ground-fault circuit interrupter:

BKW31009CK by Panasonic Electric Works, Co., Ltd.

Note the following when installing a ground-fault circuit interrupter.

- A ground-fault circuit interrupter must be installed for each power supply to the machine.
- "O" (open) or "|" (closed) must be indicated as the operation status.
- The ground-fault circuit interrupters must be equipped with an external operating mechanism (e.g., a handle) and installed 0.6 m to 1.9 m above the ground so that they can be easily accessed.
- All the power circuit phases must be shut down. However, DO NOT disconnect the ground wire or PE cable.

3. Robot Control Cable

The robot control cables must be connected by qualified personnel. Also, these processes must be compliant with the laws, regulations, and ordinances of the respective country.

Recommended protector:

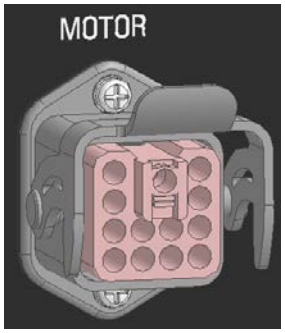
Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

⚠DANGER

- Never use unauthorized cables. Manufacturing and use of unauthorized cables can result in a fire.
- Do not insert connectors or connect to terminals that are not specified. Unspecified wiring causes fires. Also, electric shock may occur due to leakage of electricity, etc.
- Make sure to connect a ground wire. If the ground wire is not connected, there is a risk of electric shock due to leakage of electricity.
- Make sure that the connectors are securely connected. If the mating is loose, static electricity may enter and cause the robot to malfunction and collide.

3.1 Motor Drive Cable

Connect the power line to the "MOTOR" connector using the cable provided with the robot. The pin layout of the "MOTOR" connector is as follows. Connect the motor drive for the four axes, and connect the FG to the connector as a batch.



Pin No.	Signal name	Signal description	Pin No.	Signal name	Signal description
1	U1	1st axis, U phase	7	W2	2nd axis, W phase
2	V1	1st axis, V phase	8	U4	4th axis, U phase
3	U2	2nd axis, U phase	9	V3	3rd axis, V phase
4	V2	2nd axis, V phase	10	V4	4th axis, V phase
5	U3	3rd axis, U phase	11	W3	3rd axis, W phase
6	W1	1st axis, W phase	12	W4	4th axis, W phase
13	FG	Batch FG			

3.2 Brake Cable

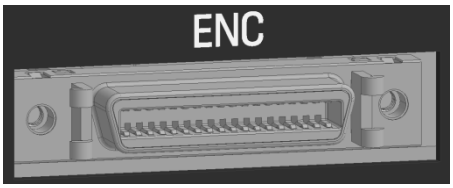
The brake line is connected to the "BRK" connector using the cable provided with the robot. The pin layout of the "BRK" connector is as follows. The brake signal line is wired to the robot's third and fourth axes.



Pin No.	Signal name	Signal description
A1	P24V	Internal P24V
B1	P24V	Internal P24V
A2	BRA3	3rd axis brake signal
B2	BRA4	4th axis brake signal
A3	P24G	Internal P24G
B3	MCONH	Servo ON LED signal
A4	P24G	Internal P24G
B4	NC	-

3.3 Encoder Cable

The encoder line is connected to the "ENC" connector using the cable provided with the robot. The pin layout of the "ENC" connector is as follows.



Pin No.	Signal name	Signal description	Pin No.	Signal name	Signal description
1	ARQ/SD	1st axis SD+	19	P5VE	5V power supply
2	ARQ/SD*	1st axis SD-	20	P5VE	5V power supply
3	DRQ/SD	2nd axis SD+	21	P5VE	5V power supply
4	DRQ/SD*	2nd axis SD-	22	P5VE	5V power supply
5	CRQ/SD	3rd axis SD+	23	P5VE	5V power supply
6	CRQ/SD*	3rd axis SD-	24	P5VE	5V power supply
7	BRQ/SD	4th axis SD+	25	P5VE	5V power supply
8	BRQ/SD*	4th axis SD-	26	P5VE	5V power supply
9	-	Reserved	27	P5VE	5V power supply
10	-	Reserved	28	P5VE	5V power supply
11	-	Reserved	29	P5VE	5V power supply
12	-	Reserved	30	P5VE	5V power supply
13	GND	P5G ground	31	GND	P5G ground
14	GND	P5G ground	32	GND	P5G ground
15	GND	P5G ground	33	GND	P5G ground
16	GND	P5G ground	34	GND	P5G ground
17	GND	P5G ground	35	GND	P5G ground
18	GND	P5G ground	36	GND	P5G ground

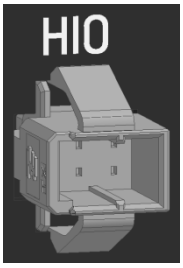
3.4 Hand I/O Cable

The TS5000-MS robot controller is connected to the robot body to control the I/O attached to the hand at the tip of the robot. Input/output 8 points/8 points can be used as standard.

- The controller and the robot are connected using the HIO connector.
- Wiring between the robot and the customer's equipment is made with the "HAND" connector on the second arm of the robot.

3.4.1 Hand I/O communication Cable

The hand I/O controller-robot connection in THE600 uses the hand I/O communication cable "HIO" connector. Connect the hand I/O communication cable with the cable attached to the robot side. The pin layout of the "HIO" connector is as follows.



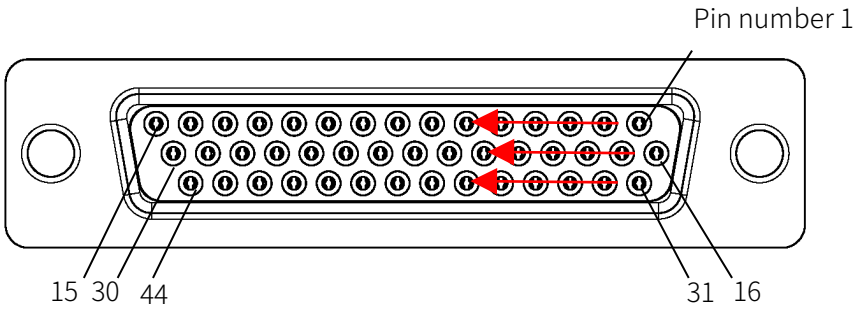
Pin No.	Signal name	Signal description
A1	EXP24V	External P24V output
B1	FUGND	External P24G ground
A2	HIO_RS485A	485 communication +
B2	HIO_RS485B	485 communication -

3.4.2 Connection Specifications between Robot and Customer Wiring

For the TS5000-MS, in the standard configuration, 8 input signals and 8 output signals are provided for controlling the end effector.

The input signal specifications are identical to those for digital input signals. (For details on the common cable connections, refer to Fig. 3.2 Connecting the Hand I/O Cable (Type N) or Fig. 3.3 Connecting the Hand I/O Communication Cable (Type P).

If the customer is using their own system for the wiring between the robot and the customer's equipment, use the hand I/O and user wiring cable (option). For details on the cables, refer to "Chapter 5.2 of the STEA1171_THE600 Robot Manual" or "Chapter8.1.1 of the STEA1245_SCARA Robot THE800/THE1000 Robot Manual". For details, please contact SHIBAURA MACHINE Customer Service or the Headquarters Sales Department.



[Pin layout by looking at the HAND connector on the robot side]

Fig. 3.1 Hand Connector Pin Arrangement

MEMO

1. *To use the hand I/O, it is necessary to connect the 24 VDC power supply from the input/output external power supply connector and the safety input/output connector (SAFE connector). For details, refer to 4.2Connecting the Input/Output External Power Supply and Safety Input/Output Cable. Also, connect the common cable as shown in Fig. 3.2 Connecting the Hand I/O Cable (Type N) or Fig. 3.3 Connecting the Hand I/O Communication Cable (Type P).*
2. *Select the most suitable external power supply to be connected according to your system specifications (current capacity).*
   As a rule of thumb, it is recommended to use about 100W since a 3A fuse is installed for I/O circuit protection.

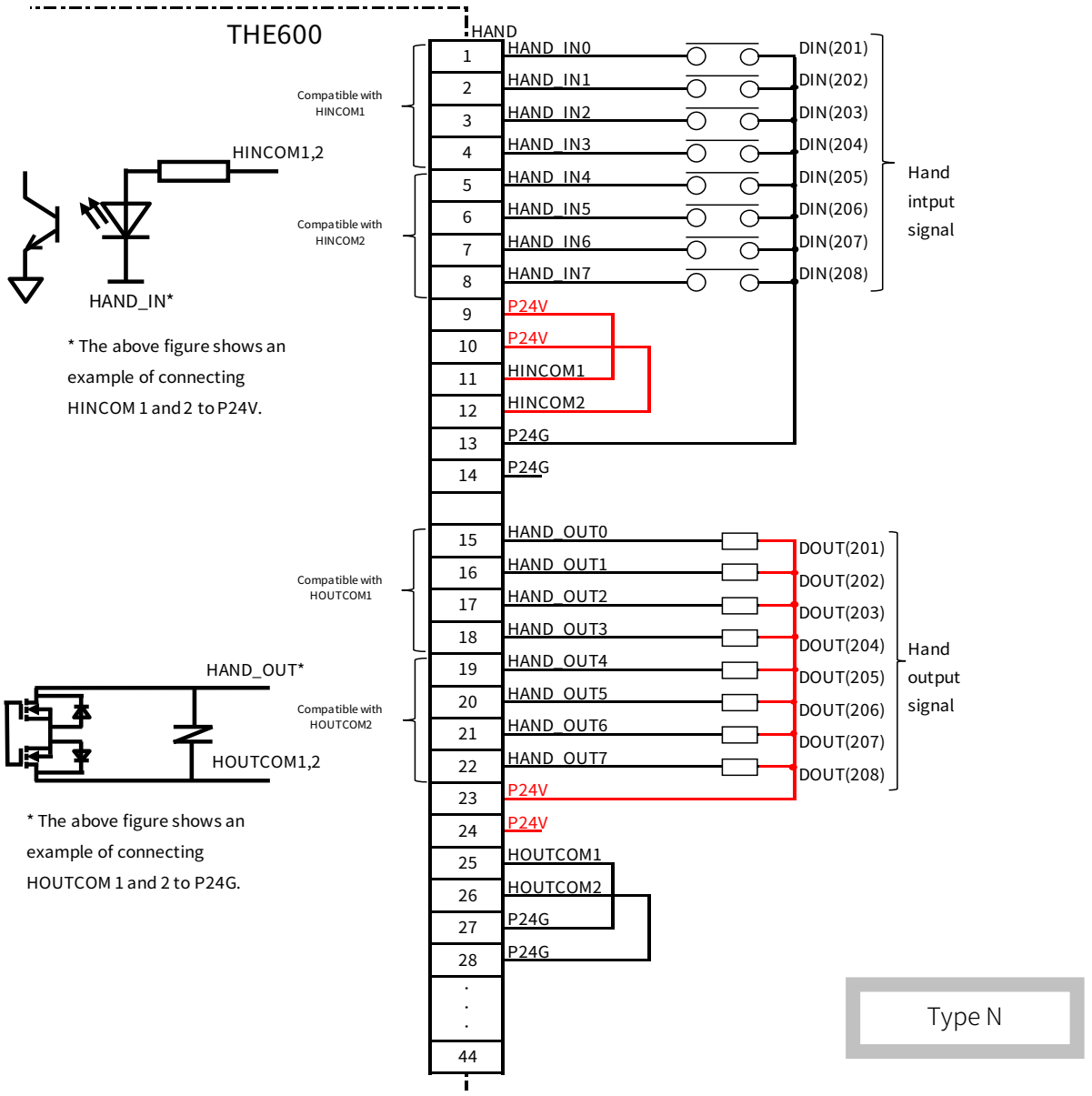
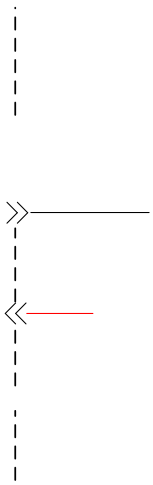


Fig. 3.2 Connecting the Hand I/O Cable (Type N)

Input method		Output specifications
Non-voltage contact specifications	Transistor open collector input	
Contact rating: 24 VDC, 10 mA or over Circuit current: Approx. 7 mA Minimum current: 24 VDC, 1 mA Contact impedance: 100 Ω or less	Withstand voltage between collector and emitter: 30 V or over Circuit current: Approx. 7 mA Current between collector and emitter: 10 mA or over Leak current between collector and emitter: 100 μA or less	Output type: Photo MOS output Electric rating: Rated voltage 24 VDC Rated current: 100 mA (MAX)/1 output

Table 3.1 Input/Output Signal Connector HAND (Type-N)

HAND pin number	HAND signal name		Signal number DIN/DOUT	Input/output circuit/connection example
1	HAND_IN0	Input signal 1	DIN(201)	Input 
2	HAND_IN1	Input signal 2	DIN(202)	
3	HAND_IN2	Input signal 3	DIN(203)	
4	HAND_IN3	Input signal 4	DIN(204)	
5	HAND_IN4	Input signal 5	DIN(205)	
6	HAND_IN5	Input signal 6	DIN(206)	
7	HAND_IN6	Input signal 7	DIN(207)	
8	HAND_IN7	Input signal 8	DIN(208)	
9	P24V			
10				
11	HINCOM1			
12	HINCOM2			
13	P24G			
14				
15	HAND_OUT0	Output signal 1	DOUT(201)	Output 
16	HAND_OUT1	Output signal 2	DOUT(202)	
17	HAND_OUT2	Output signal 3	DOUT(203)	
18	HAND_OUT3	Output signal 4	DOUT(204)	
19	HAND_OUT4	Output signal 5	DOUT(205)	
20	HAND_OUT5	Output signal 6	DOUT(206)	
21	HAND_OUT6	Output signal 7	DOUT(207)	
22	HAND_OUT7	Output signal 8	DOUT(208)	
23	P24V			
24				
25	HOUTCOM1			
26	HOUTCOM2			
27	P24G			
28				
~	User wiring			
44				

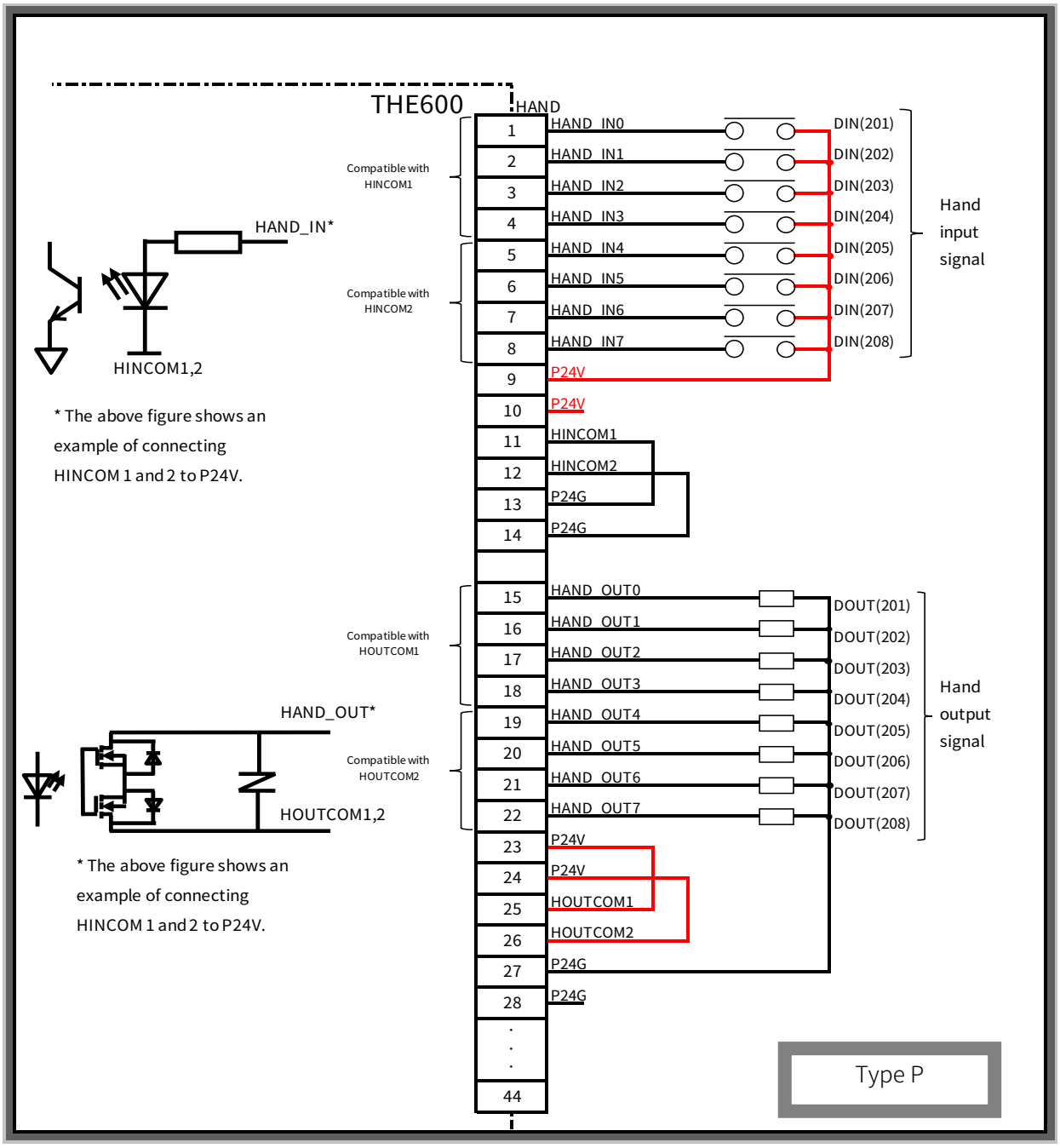
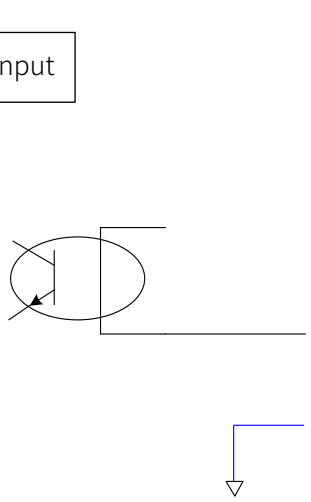


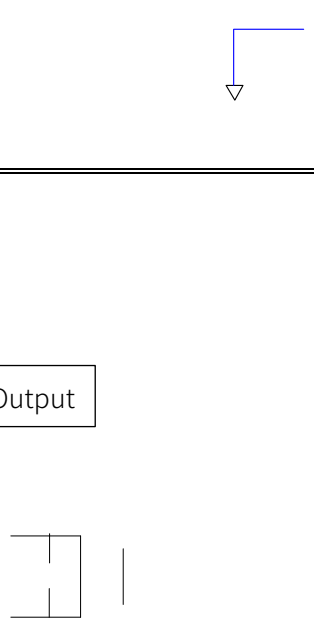
Fig. 3.3 Connecting the Hand I/O Communication Cable (Type P)

Input method		Output specifications
Non-voltage contact specifications	Transistor open collector input	
Contact rating: 24 VDC, 10 mA or over Circuit current: Approx. 7 mA Minimum current: 24 VDC, 1 mA Contact impedance: 100 Ω or less	Withstand voltage between collector and emitter: 30 V or over Circuit current: Approx. 7 mA Current between collector and emitter: 10 mA or over Leak current between collector and emitter: 100 μA or less	Output type: Photo MOS output Electric rating: Rated voltage 24 VDC Rated current: 100 mA (MAX)/1 output

Table 3.2 Input/Output Signal Connector HAND (Type-P)

HAND pin number	HAND signal name		Signal number DIN/DOUT	Input/output circuit/connection example
1	HAND_IN0	Input signal 1	DIN(201)	
2	HAND_IN1	Input signal 2	DI(202)	
3	HAND_IN2	Input signal 3	DIN(203)	
4	HAND_IN3	Input signal 4	DIN(204)	
5	HAND_IN4	Input signal 5	DIN(205)	
6	HAND_IN5	Input signal 6	DIN(206)	
7	HAND_IN6	Input signal 7	DIN(207)	
8	HAND_IN7	Input signal 8	DIN(208)	
9	P24V			
10				
11	HINCOM1			
12	HINCOM2			
13	P24G			
14				
15	HAND_OUT0	Output signal 1	DOUT(201)	
16	HAND_OUT1	Output signal 2	DOUT(202)	
17	HAND_OUT2	Output signal 3	DOUT(203)	
18	HAND_OUT3	Output signal 4	DOUT(204)	
19	HAND_OUT4	Output signal 5	DOUT(205)	
20	HAND_OUT5	Output signal 6	DOUT(206)	
21	HAND_OUT6	Output signal 7	DOUT(207)	
22	HAND_OUT7	Output signal 8	DOUT(208)	
23	P24V			
24				
25	HOUTCOM1			
26	HOUTCOM2			
27	P24G			
28				
~	User wiring			
44				





3.4.3 AUX Settings of Hand Output Signals

The hand output signals can be turned ON or OFF manually from the teach pendant. However, when turning the hand output signals ON/OFF using AUX, they must be defined in "USER.PAR" (user parameter). For the settings, refer to “STEA1166_Parameter Manual [U018] AUX SIGNAL”.

MEMO

1. Two hand output signals can be combined to enable control of a double solenoid device.
In this case, the two output signals with a continuous signal name are used for the double solenoid. Write the program in the robot language so that the two output signals of the double solenoid have an exclusive OR relationship during automatic operation. Also, be aware that the two output signals turn off when the controller power is shut off or when the power is turned on.

2. In the OPEN1, OPEN2, CLOSE1, CLOSE2, OPENI1, OPENI2, CLOSEI1, and CLOSEI2 commands, each (201, 202) and (203, 204) signal group must be output using an exclusive OR operation.

3. The hand output signals are not reset by the "RESET SIG" operation or "RESET DOUT" command.

NOTICE

- The hand output signal retains its current status when AC 200V is turned off while external 24V is still supplied. When AC 200V is turned on again, the hand output is cleared. Therefore, set the robot's hand output to turn off when closing the hand and turn on when opening the hand, or implement exclusive control using a double solenoid.

4. Input/Output External Power Cable and Safety Input/Output Cable

The input/output external power supply and safety input/output cable must be connected by qualified personnel. Also, these processes must be compliant with the laws, regulations, and ordinances of the respective country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

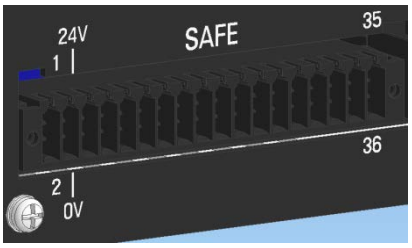
⚠DANGER

- Never use unauthorized cables. Manufacturing and use of unauthorized cables can result in a fire.
- Do not insert connectors or connect to terminals that are not specified. Unspecified wiring causes fires. Also, electric shock may occur due to leakage of electricity, etc.
- Make sure to connect a ground wire. If the ground wire is not connected, there is a risk of electric shock due to leakage of electricity.
- Make sure that the connectors are securely connected. If the mating is loose, static electricity may enter and cause the robot to malfunction and collide.

4.1 Input/Output External Power Supply Connector and Safety Input/Output Connector (SAFE Connector)

If used in the customer's own system, connect the cable to the supplied "SAFE" connector [B2CF 3.50/36/180LR (plug type)] (accessory) on the robot controller side. This is used as an input/output connector for contact input for the external emergency stop switch, safety door, and other components in the upper-level system and for relay outputs that indicate the controller status. Pins 3 and 4 of this signal are used to supply an external 24 V.

Board side connector: S2C-SMT 3.50/36/90LF * Weidmüller Interface GmbH & Co. KG



Pin No.	Signal name	Signal description	Pin No.	Signal name	Signal description
1	-	Unused	19	-	Unused
2	-	Unused	20	-	Unused
3	EXP24V	External 24V input	21	-	Unused
4	PG	P24V ground	22	-	Unused
5	-	Unused	23	-	Unused
6	-	Unused	24	-	Unused
7	-	Unused	25	ENA11	Safety door input 1-1
8	-	Unused	26	ENA12	Safety door input 1-2
9	EMG11	External emergency stop input 1-1	27	ENA21	Safety door input 2-1
10	EMG12	External emergency stop input 1-2	28	ENA22	Safety door input 2-2
11	EMG21	External emergency stop input 2-1	29	-	Unused
12	EMG22	External emergency stop input 2-2	30	-	Unused
13	-	Unused	31	-	Unused
14	-	Unused	32	-	Unused
15	-	Unused	33	ENAROUT	Safety door status output
16	-	Unused	34	FUGND	External P24G ground
17	EMGROUT	Emergency stop status output	35	SVST_A	Servp ON status contact
18	FUGND	External P24G ground	36	SVST_B	Servo ON status contact

4.2 Connecting the Input/Output External Power Supply and Safety Input/Output Cable

The standard input/output external power supply and safety input/output cable connections are as follows. Emergency stop signal input and safety door signal input can be wired to the robot controller from the outside.

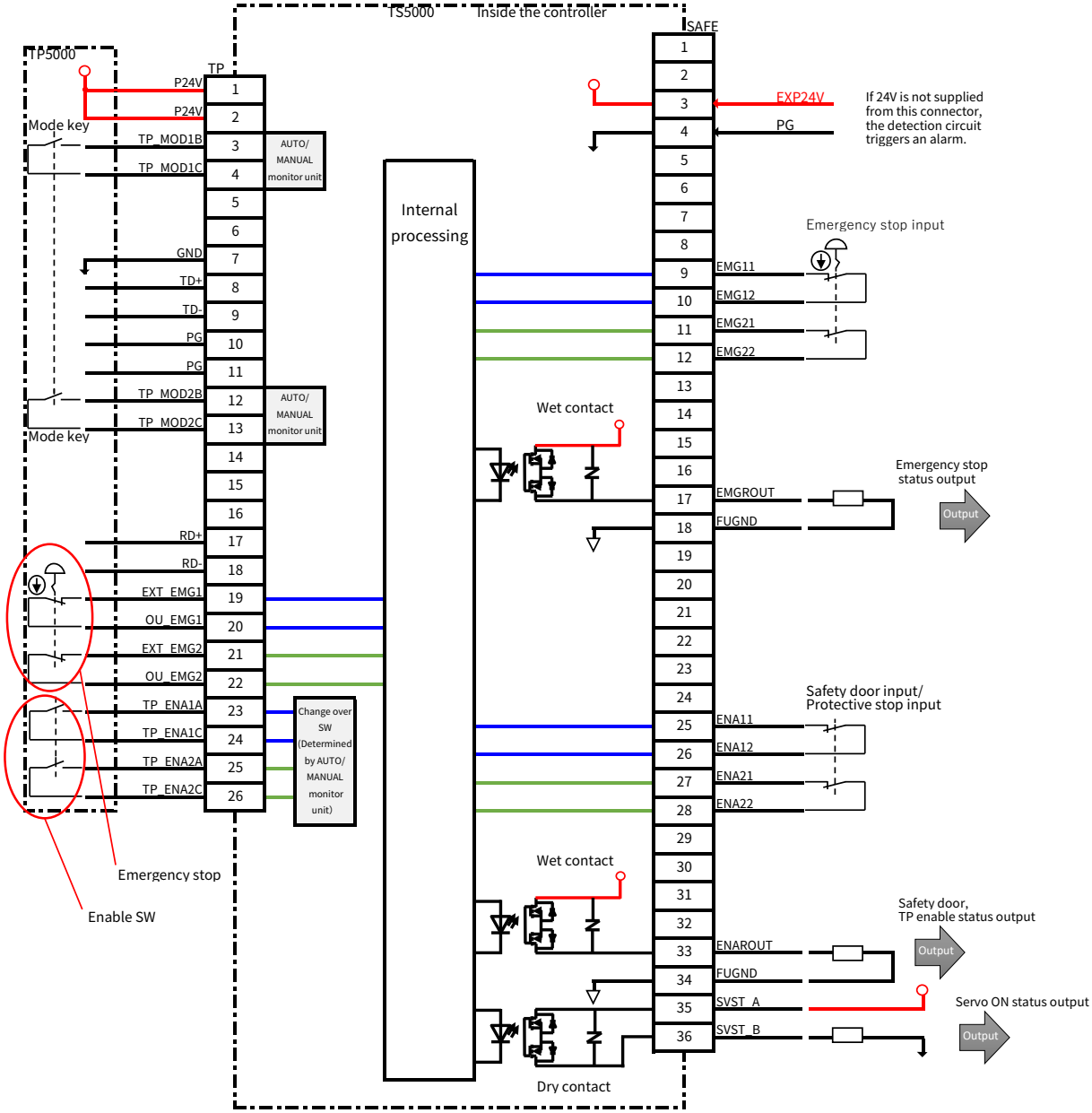


Fig. 4.1 SAFE Connector Location (Standard)

Emergency stop status output: EMGROUT/ Safety door status output: ENAROUT	Servo ON status contact: SVST
Output specification: Wet contact Output type：Photo MOS output Rated voltage：DC24V(supplied by EXP24V) Rated current：100mA(MAX)/1 output	Output specification: Dry contact Output type：Photo MOS output Rated voltage：DC24V Rated current：1.0A(MAX)/1 output

4.3 SAFE Connector Pin Details

4.3.1 External input/output power cable

・SAFE pin number 3,4

Select a power supply that has an input voltage range of 24 VDC±10% and that is SELV (Safety Extra-Low Voltage). (UL60950 Certified Product)

The I/O that is operated by an external power supply (24 VDC) is as follows.

- Safety circuits
- External input/output
- External operation input/output
- Hand input/output
- Extension input/output

Select the optimum external power supply based on customer's system specifications (current capacity).

Table 4.1 I/O External Power Supply Specifications

Power supply	DC 24 V±10%
Supported cables	0.14mm^2 to 1.5mm^2 (AWG26 to 16)

NOTICE

- Apply a voltage in the range of 24 V±10% to the external input/output power cable. Applying a voltage that exceeds the allowable voltage could damage the controller.

4.3.2 External emergency stop input 1,2 EMG

・SAFE pin number 9,10,11,12

Use an emergency stop switch that complies with the standards. This signal is emergency stop of stop category 1.

An emergency stop can be applied to robot operation from an external device. This signal is open and performs the urgent emergency stop process on the robot. If either of these signals is open, the emergency stop status cannot be released. When an emergency stop input contact is open, the emergency stop status output "EMGROUT" becomes Low. When an emergency stop input contact is close, the emergency stop status output "EMGROUT" becomes High.

MEMO

1. While this signal is open, the servo power cannot be turned on in any mode.

*2. EMG*1 to EMG *2 are assumed to have two (2) b-contacts interconnected, which should be turned on and off at the same time. If there is a delay in contact input operation, the non-conformity detection function of hardware works to retain emergency stop.*

If error clear is not done even after emergency stop latch input, confirm whether this signal has two (2) b inputs interconnected.

3. Unless this signal is used, short-circuit SAFE-9 to SAFE-10 and SAFE-11 to SAFE-12.

4. Choose an emergency stop device which has such a structure that when the emergency stop device works, the circuit will not be automatically reset until the emergency stop device is reset manually and the contact will be reset only after the emergency stop device is reset.

5. Choose any of the following shapes of emergency stop device that is suitable for the machine structure and characteristics so that unsafe conditions can be eliminated:

1) Mushroom-type pushbutton

2) Rope pull type, or bar type

3) Belly or knee action type

4) Pedal switch without a protective cover

6. The pushbutton type of emergency stop device shall have a red actuator on a yellow background.

7. The rope pull type of emergency stop device shall always maintain proper tension on the rope, and the rope shall be identified colored red and yellow.

8. In order to check the operation of the emergency stop switch after the initial startup and system configuration, check if "008-014 Emergency Stop SW ON" is displayed on the alarm screen of the teach pendant when the switch is turned on.

4.3.3 Emergency stop status output EMGROUT

・SAFE pin number 17

A P24V voltage is output from this signal when the external emergency stop 1, 2 and the emergency stop switch of the teach pendant are all closed.

This signal can always be used regardless of the mode selected by means of the master mode.

MEMO

1. This signal is not a safety contact output.

2. This contact is wet contact (24V).

2. Voltage will not be output unless power (P24V) is input to SAFE connectors 3 and 4.

3. The current capacity is 100mA (MAX).

4.3.4 Safety door input 1,2 ENA

・SAFE pin number 25,26,27,28

Use a safety door that complies with the standards. This signal is protective stop of stop category 1.

Used to emergency-stop the robot from the external equipment. While this signal is open, the processing of robot emergency stop is executed. When the safety door input contact is close, the safety door status output " ENAROUT " becomes Low.

When the safety door input contact is open, the safety door status output " ENAROUT " becomes High.

MEMO

1. While this signal is open, the servo power cannot be turned on in any mode.

*2.“ENA*1 to ENA*2”are assumed to have two (2) b contacts interconnected, which should be turned on and off at the same time. If there is a delay in contact input operation, the non-conformity detection function of hardware works to retain emergency stop. If error clear is not done even after emergency stop latch input, confirm whether this signal has two (2) b inputs interconnected.*

3. In order to check the operation of the safety door after the initial startup and system configuration, check whether “008-017 Safety SW ON” is displayed on the alarm screen of the teach pendant when the safety door contact is open.

4.3.5 Safety door status output ENAROUT

・SAFE pin number 29, 30, 31, 32

In Auto mode, it is shown in the table below.

・Auto

Safety door Input 1	Safety door Input 2	Safety door status output
Close	Close	High(24V)
Open	Close	Low
Close	Open	Low

In Manual mode, the enable switch of the teach pendant is enabled and the output is as shown in the table below.

・Manual

Safety door Input 1	Safety door Input 2	Teach pendant Enable switch	Safety door status output
Close	Close	Close	High(24V)
Open	Close	Close	Low
Close	Open	Close	Low
Close	Close	Open	Low

MEMO

1. This signal is not a safety contact output.

2. This contact is wet contact (24V).

3. There is no voltage output unless power (P24V) is input to SAFE connectors 3 and 4.

4. The current capacity is 100mA (MAX).

4.3.6 Servo ON status contact SVST

・SAFE pin number 35,36

While the servo power is turned on, the output terminals are short-circuited.

This signal can always be used regardless of the mode selected by means of the master mode.

MEMO

1. The signal is of a non-voltage contact output type. The logic is a a-contact output.

2. The contact capacity is 240 VDC (max.), 1.0 A (max.).

3. It takes about three (3) seconds from the servo ON to the time when the robot is actually ready to work. Thus, it is necessary to program a delay of 3.5 seconds or over between the SV_RDY signal ON and the RUN signal ON.

4.4 Enable switch state depending on the running MODE

The operation of this signal from the teach pendant varies depending on the mode selected by means of the master mode switch on the controller front panel.
In MANUAL mode, the enable switch on the teach pendant becomes enabled, and enters close state by gripping the switch to neutral.
In AUTO mode, the enable switch on the teach pendant becomes disabled, but the safety door switch, etc. is enabled.

The operational correspondence is shown below.

	Teach pendant enable switch	Safety door input	Emergency stop input	Speed limit
MANUAL mode	Enabled	Enabled	Enabled	20%
AUTO mode	Disabled	Enabled	Enabled	25% or 100%

MEMO

1. In AUTO mode, when the low speed signal (LOW_SP) of the INPUT connector is "L", the speed is limited to 25%.

4.5 Example of emergency stop line connection for two controllers

The figure below shows an example of emergency stop line connection. The emergency stop input of TS5000-MS comes from pin numbers 9 - 12 of safe connector (4.3.2External emergency stop input 1,2 EMG) and emergency stop button of TP5000.

Since the emergency stop status is output as a wet contact from pin numbers 17 and 18 of the SAFE connector(4.3.3Emergency stop status output EMGROUT), the two controllers can be interconnected for emergency stop circuit integration by wiring them as shown in the figure below.

It is also recommended to monitor the status of the safety relay using a b-contact paired with an a-contact.

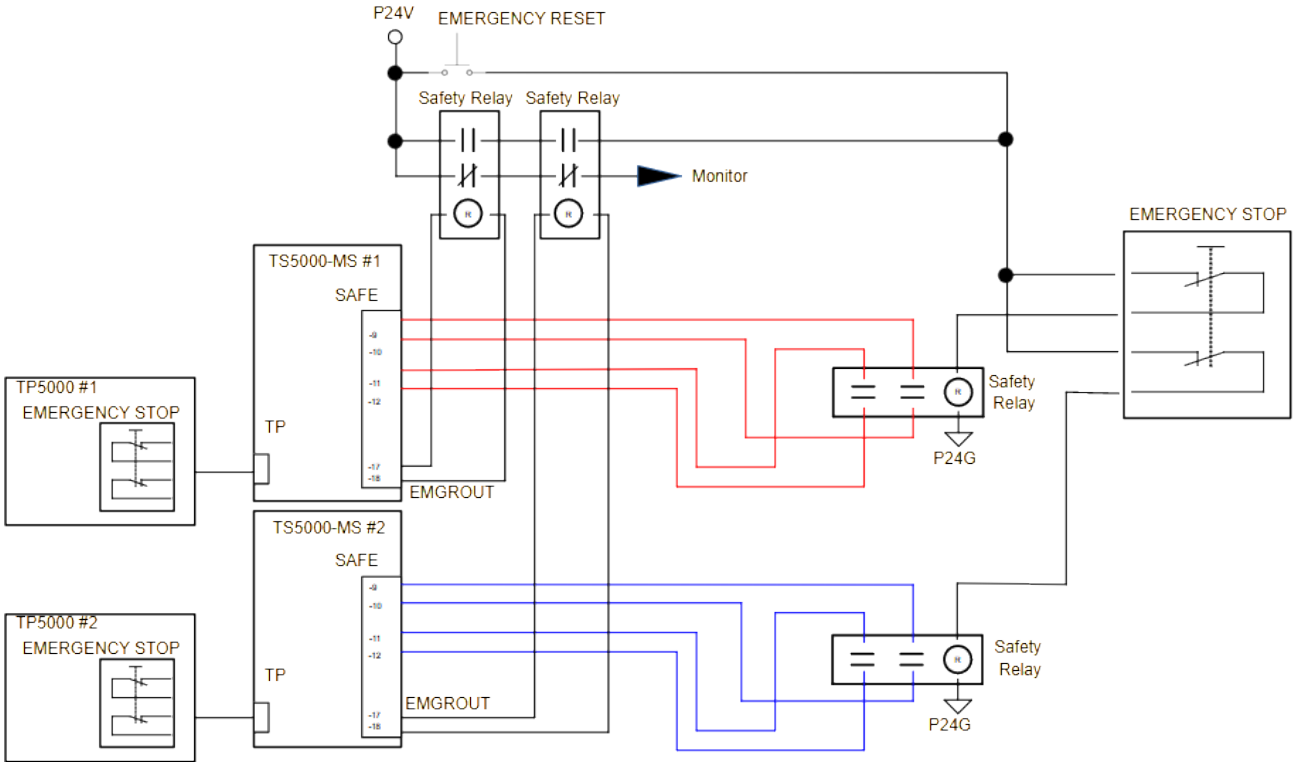


Fig. 4.2 Example of emergency Stop Line Connection

MEMO

1. *I In this case, all connected functions become emergency stop status when the emergency button is pressed. It is necessary to input an emergency stop reset after the emergency stop contact is closed in order to release the emergency stop.*
2. *When the power is off, the system becomes an emergency stop.*
3. *When connecting multiple devices, use a safety PLC or safety controller.*

4.6 Fabricating the Cable for the Connector for Input/Output External Power Supply and Safety Input/Output

Follow the points below when fabricating the external I/O signal cable.

a.) Use the supplied connector for the SAFE connector of the robot controller.
The connector's withstanding voltage is 300V and the maximum load is 9.5 A.
For more information on the contact capacity of each signal, see 4.3 SAFE Connector Pin Details.

Connector name	Type of connector	Maker
SAFE connector	B2CF 3.50/36/180LR	Weidmüller Interface GmbH & Co. KG

b.) Select the cable to be used based on the specifications below.

- Core wires: 0.14 mm^2 to 1.5 mm^2 (AWG26 to AWG16) twisted wires, single wires having withstanding voltage 300 V/9.5 A or less
- Cable length: 30 m max.

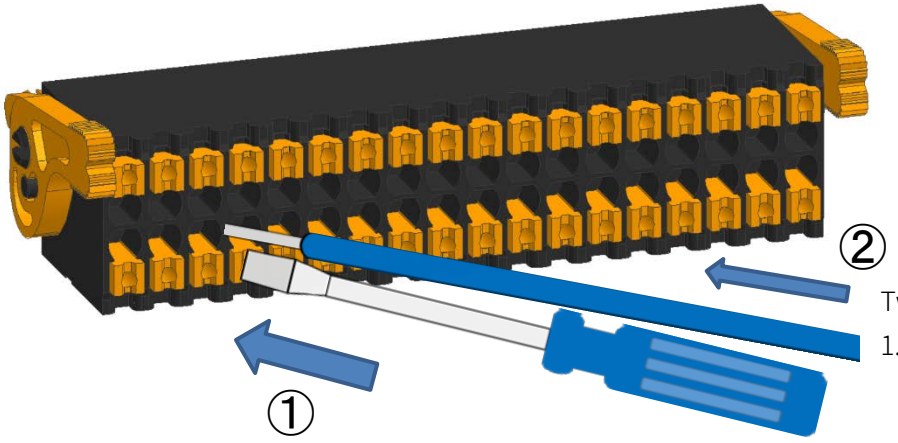
⚠DANGER

- Never use wires other than those specified. Using a cable that differs from the specifications may cause a fire due to short circuit or overheating of the core wire. Also, the robot may malfunction due to noise intrusion and other factors, and may be struck hard.

NOTICE

- For more information on the contact capacity of each signal, see 4.3 SAFE Connector Pin Details. Applying a voltage or current that exceeds the contact capacity will damage the controller.

c.) Four SAFE jumper cables are supplied for the controller. This cable can be inserted directly into the connector for easily enabling a jumper connection. If the jumper connection is performed for models with standard specifications, refer to “STEA1161_Controller Manual 4.1.4 External power supply for input/output and connection of connectors for safety input/output "SAFE"”.
If the SAFE jumper cable will not be used, first, push a slotted screwdriver into the orange section as shown in the figure below, and then insert the wire.



Twisted cable 0.14mm² to 1.5mm²

Slotted screwdriver

d.) Controller side pin arrangement

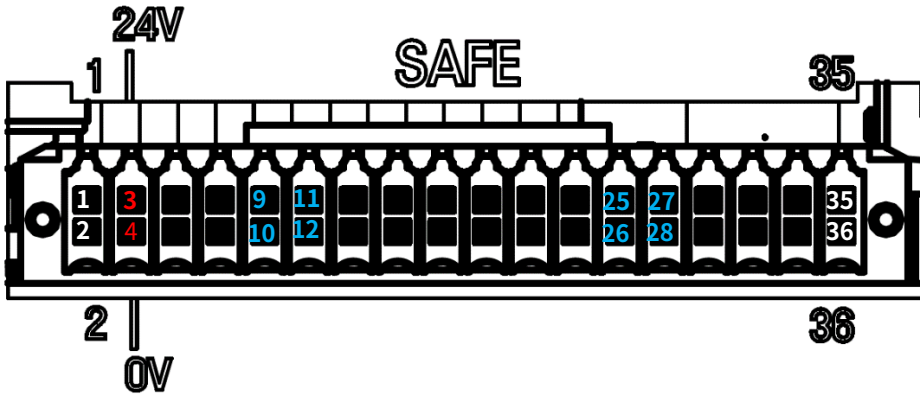


Fig. 4.3 SAFE Connector Pin Arrangement

e.) Simple wiring of SAFE connector

If no emergency stop switch or safety door is installed externally, wire as shown in Fig. 4.4.

In order to operate the input/output signals, it is necessary to connect the external 24V to pin 3 and the 0V side of external 24V to pin 4.

Also, if there is no connection between the emergency stop lines of 9-10 pins and 11-12 pins and the safety door input lines of 25-26 pins and 27-28 pins, the robot will stop.

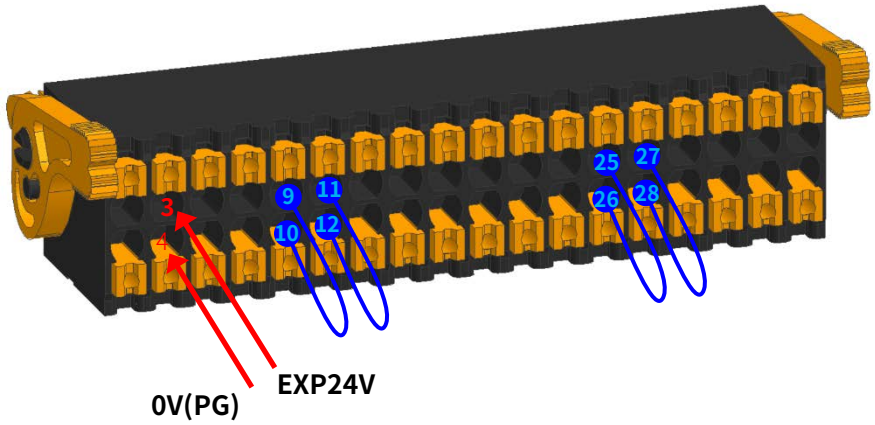


Fig. 4.4 Jumper Method of Standard Specifications

4.7 Attaching and Detaching SAFE Connector

Before attaching or detaching the SAFE connector, be sure to always shut off the main power in the control panel of the customer. Also, check that the external power supply (24 VDC) is shut off.

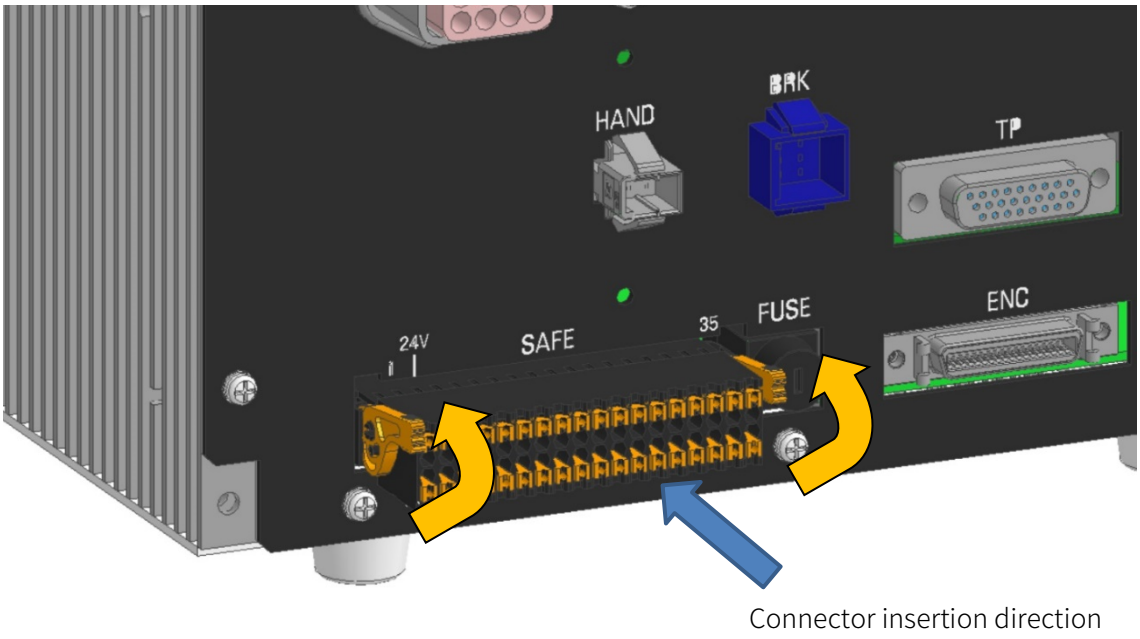


Fig. 4.5 Attaching and Detaching SAFE Connector

To connect the SAFE connector, fully insert the cable-side connector into the controller-side connector, and then use the levers to lock both ends of the connector in place. To remove, in the opposite way from connecting, release the levers on both ends of the connector, and then pull out the connector.

⚠ WARNING

- When disconnecting cables, grasp the connector while holding the controller, and pull the connector to remove it.
  - Pulling the cable itself could break the cable wires, resulting in an electric shock.

NOTICE

- Turn off the power before connecting and disconnecting cables from the controller. If you touch the terminal of the cable that you removed while the power is on, you will get an electric shock. Also, if the cable is connected or disconnected while the main power is on, the IC on the board will be damaged.
- When disconnecting cables, grasp the connector while holding the controller, and pull the connector to remove it.
  - The controller can flip over and be damaged.

5. Input Signal Cable

The cables must be connected by qualified personnel. Also, these processes must be compliant with the laws, regulations, and ordinances of the respective country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

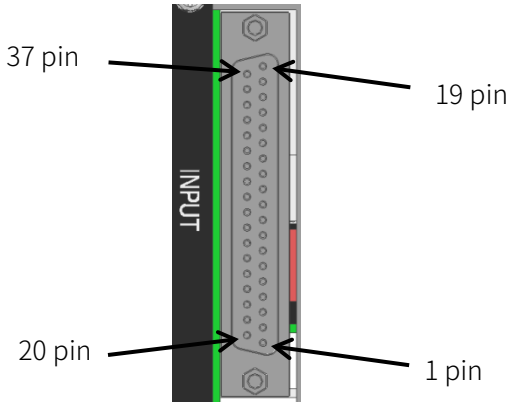
⚠DANGER

- Never use unauthorized cables. Manufacturing and use of unauthorized cables can result in a fire.
- Do not insert connectors or connect to terminals that are not specified. Unspecified wiring causes fires. Also, electric shock may occur due to leakage of electricity, etc.
- Make sure to connect a ground wire. If the ground wire is not connected, there is a risk of electric shock due to leakage of electricity.
- Make sure that the connectors are securely connected. If the mating is loose, static electricity may enter and cause the robot to malfunction and collide.

5.1 Connecting the Input Signal Cable

The input signal cable is an interface that recognizes signals input to the robot controller from an external device. The input signals include 8 general-purpose input signals that can be used by the user for programming using SCOL2 language or the sequence function in addition to 13 system input signals that allow use of the controller from an external device. In AUTO mode, the robot can be controlled by inputting system input signals, such as the servo ON signal, from a sequencer or other device.
Also, when the servo ON or other signals are not required from a sequencer, etc., the robot can be operated without special wiring by connecting the INPUT jumper connector.
The external input signal cable is connected using the supplied dummy connectors [XM3A-3721 (plug type), XM2S-3711 (connector cover)]. In these dummy connectors, some of the system input signals (b-contact configuration section) use jumper connections. If necessary, the customer can use these signals by removing these jumper connectors. Cables with one side soldered to the connector are available as an option. Please contact SHIBAURA MACHINE Customer Service or Sales Department at the headquarters.
For details about cables, refer to13.1 INPUT Cable.

Board-side connector:
XM3B-3722-112 * OMRON Corporation



Pin No.	Signal name	Signal description	Pin No.	Signal name	Signal description
1	INPUT0	General-purpose input 1	20	INPUT2	General-purpose input 2
2	INPUT2	General-purpose input 3	21	INPUT4	General-purpose input 4
3	INPUT4	General-purpose input 5	22	INPUT6	General-purpose input 6
4	INPUT6	General-purpose input 7	23	INPUT8	General-purpose input 8
5	EXP24V	External P24V power supply	24	FUGND	External P24G ground
6	INCOM	Input common	25	STROBE	Strobe
7	PRG_RST	Program reset	26	STEP_RST	Step reset
8	CYC_RST	Cycle reset	27	DO_RST	Command reset
9	ALM_RST	Alarm reset	28	RUN	Run
10	EX_SVON	External servo ON	29	EXP24V	External P24V power supply
11	FUGND	External P24G ground	30	SYSINCOM	Input common
12	STOP	Stop	31	CYCLE	Cycle operation
13	LOW_SP	Speed limit	32	BREAK	Pause
14	SVOFF	Servo OFF	33	NC	-
15	NC	-	34	NC	-
16	EXP24V	External P24V power supply	35	FUGND	External P24G ground
17	SYSINCOM	Input common	36	NC	-
18	NC	-	37	NC	-
19	NC	-			

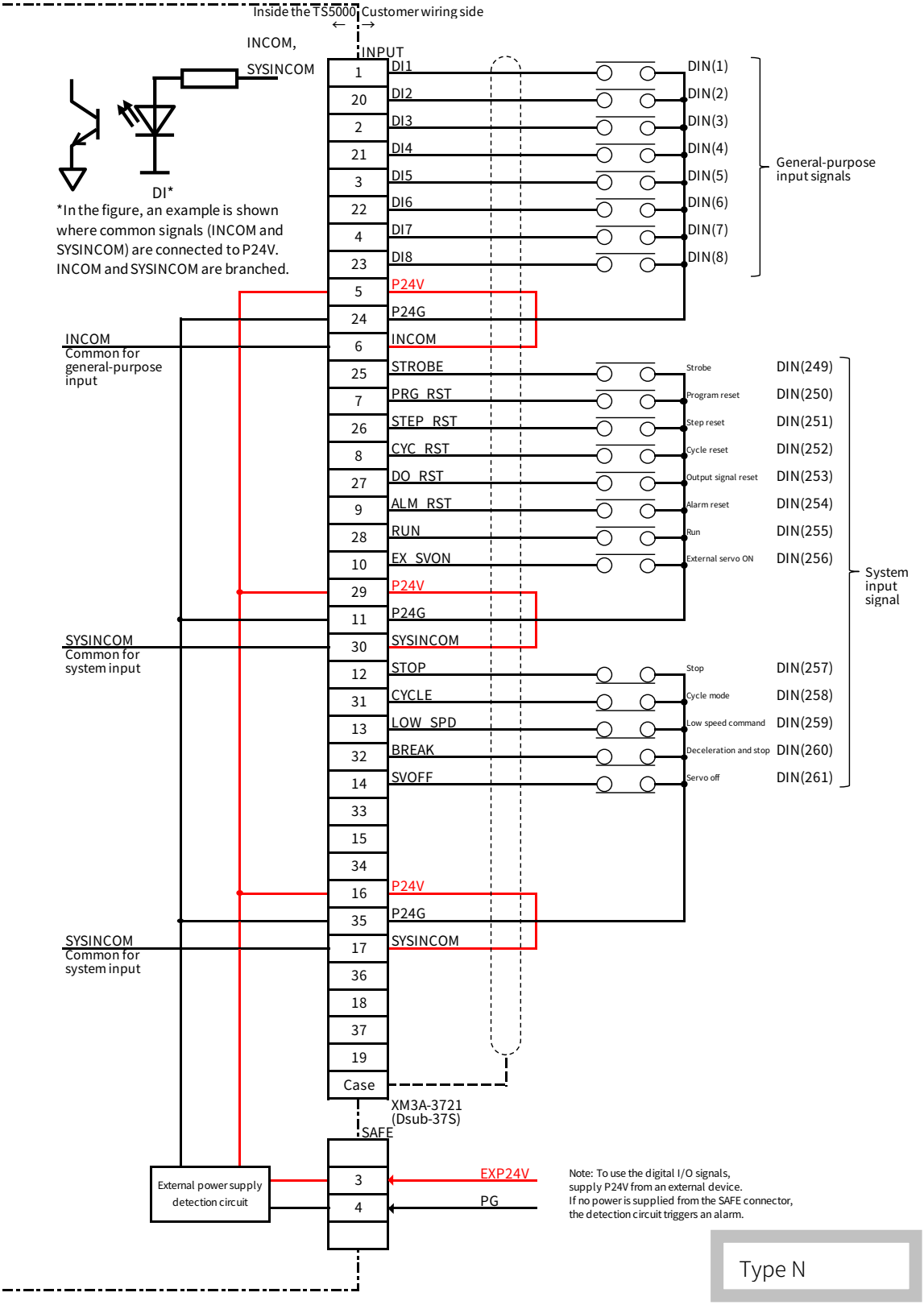


Fig. 5.1 Connecting the Input Signal Cable (Type N)

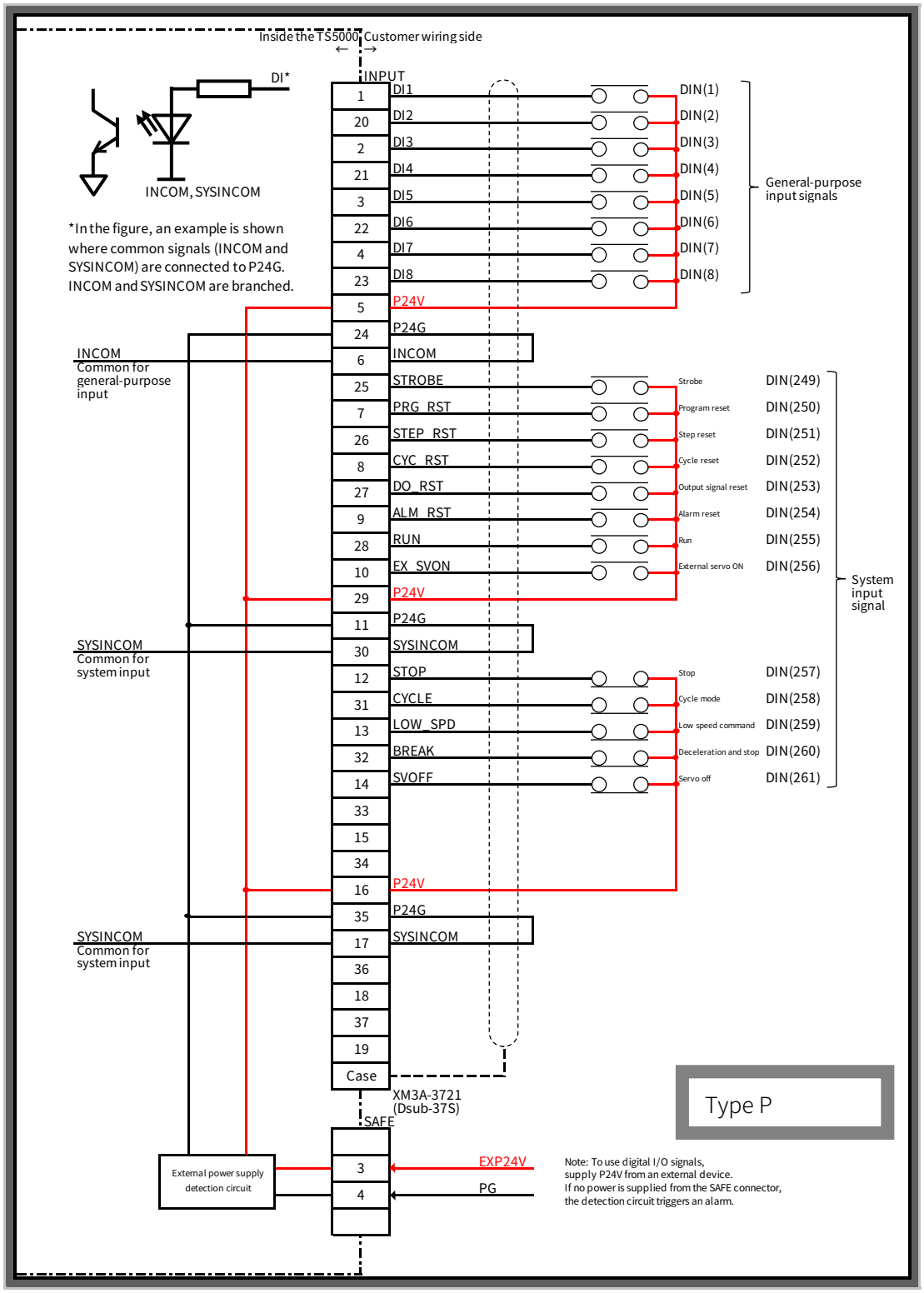


Fig. 5.2 Connecting the Input Signal Cable (Type P)

MEMO

1. *The input signals are non-voltage contact inputs or open collector inputs, and except for some of the system inputs (STOP, CYCLE, LOW_SPD, BREAK, SVOFF) that are b-contact input, all are a-contact input.*
2. *For more information on the functions, circuits used, common line changeover, and other details for each signal, refer to the following pages.*

5.2 General-purpose input signals DIN1 to DIN8

・INPUT pin number 1 to 4, 20 to 23

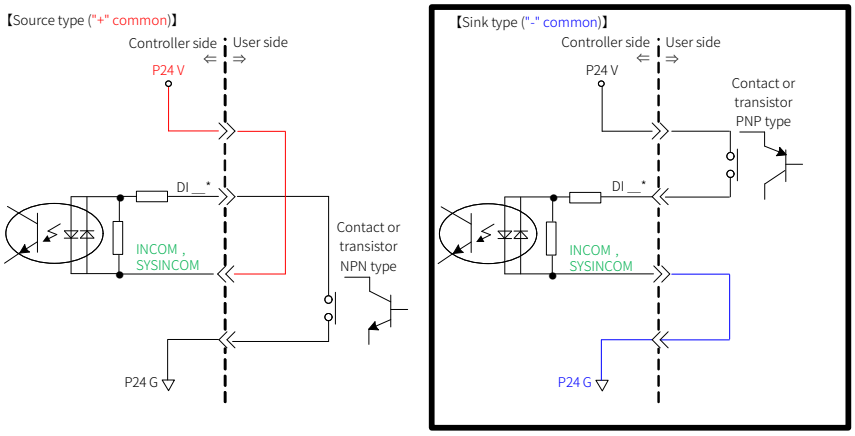
DIN1 to DIN8 of the COM line are INCOM.

Each signal state of DIN1 to DIN8 can be identified by the robot program (DIN command) to branch the processing of program. Also, it is possible to perform interruptive processing of each signal (DIN1 to DIN8) with change in signal state monitored during the robot operation.

Specifications of non-voltage contact and transistor

Non-voltage contact specifications	Transistor specifications
Contact rating 24 VDC, 10 mA or over Circuit current Approx. 7 mA Minimum current 24 VDC, 1 mA Contact impedance 100 Ω or less	Withstand voltage between collector and emitter 30 V or over Circuit current: Approx. 7 mA Current between collector and emitter 10 mA or over Leakage current between collector and emitter 100 μA or less

Refer to the following graphs for details on the connection method (example of used circuit).



MEMO

1. When the pulse type input signals are used, the pulse width should be 100 ms or over.

5.3 System Input Signal

・INPUT pin number 7 to 10, 12 to 14, 25 to 28,31 to 32

The system input signals of the COM line are SYSINCOM.

The system input signals are a total of 13 signals that control STOP, CYCLE, and other commands from an external device. An exclusive input terminal is set for each function.

The structure of the system input signal is quite the same as that of the general-purpose input signal. See 5.2 General-purpose input signals DIN1 to DIN8.

When inputting the system input signal, keep the input state until the output signal corresponding to each input is output to assure each signal input. The relationship between input signal and output signal is stipulated by the timing chart of each signal. Also, the system input signals include those that are enabled and disabled based on the controller master mode.

For details on the signals that are enabled, refer to Table 5.1.

Table 5.1 List of system input signal ON modes

Master mode switch / Designation	Master mode	
	AUTO	MANUAL
STROBE (Strobe)	○	
PRG_RST (Program reset)	○	
STEP_RST (Step reset)	○	
CYC_RST (Cycle reset)	○	
DO_RST (Output signal reset)	○	
ALM_RST (Alarm reset)	○	
RUN (Start)	○	
EX_SVON (External servo ON)	○	
STOP (Stop)	○	○
CYCLE (Cycle operation mode)	○	
LOW_SPD (Low speed command)	○	○
BREAK (Deceleration and stop)	○	○
SVOFF (Servo OFF)	○	○

○ : Available

⚠ DANGER

- Mount a cover or other enclosure on the buttons and switches on the control device connected to the controller. If a signal is turned on due to unintentional contact, the robot could be activated, resulting in a collision.
- Display the operation mode of each controller on the operation device. If you are unaware of the operation mode, a collision could occur with a robot that is performing an unexpected operation.

Table 5.2 List of the system input signals

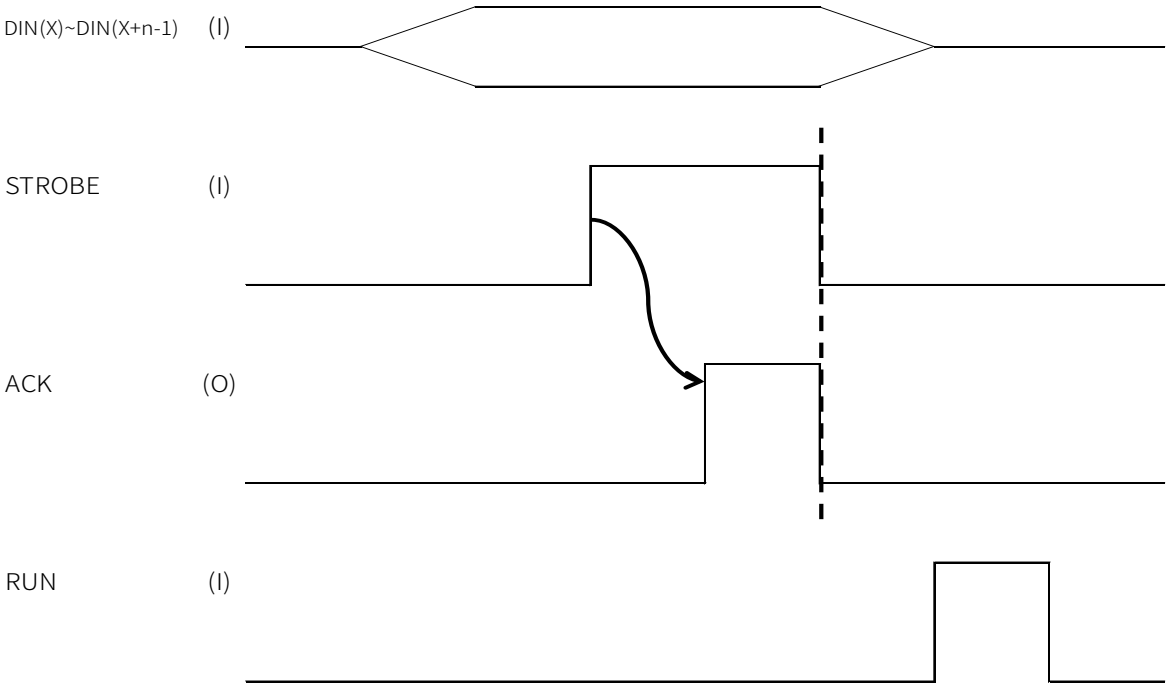
Name	Signal name in DIN command	Function	Signal judgment
STROBE	249	Selects an execution program file from the external equipment.	ON: Short-circuit OFF: Open
PRG_RST	250	Resets a program currently stopped to step 1. The value of each variable is also reset to "0".	ON: Short-circuit OFF: Open
STEP_RST	251	Resets a program currently stopped to step 1. The value of each variable remains unchanged.	ON: Short-circuit OFF: Open
CYC_RST	252	Resets a program currently stopped to the step labeled "RCYCLE". The value of each variable remains unchanged.	ON: Short-circuit OFF: Open
DO_RST	253	Batch-resets the digital output signals (DO_1 to DO_64, DO_101 to DO_164, DO_301 to DO_2264).	ON: Short-circuit OFF: Open
ALM_RST	254	Cancels an alarm which has occurred while the robot controller was able to work.	ON: Short-circuit OFF: Open
RUN	255	Starts a program currently stopped to execute an automatic operation.	ON: Short-circuit OFF: Open
EX_SVON	256	Turns on the servo power.	ON: Short-circuit OFF: Open
STOP	257	Stops a program during automatic operation.	ON: Short-circuit OFF: Open
CYCLE	258	Stops a program currently executed in automatic operation after one (1) cycle has finished.	ON: Open OFF: Short-circuit
LOW_SPD	259	Reduces the robot operating speed in automatic operation.	ON: Open OFF: Short-circuit
BREAK	260	Slows down and stops the ongoing robot operation.	ON: Open OFF: Short-circuit
SVOFF	261	Turns off the servo power.	ON: Open OFF: Short-circuit

5.3.1 STROBE (Strobe) Signal

・INPUT pin number 25

STROBE is an input signal for enabling the EXTSELECT function. The EXTSELECT function can be used to select an executable file using input signals.
The program number selected should use any successive "n" numbers (max. eight (8) numbers) of external digital input signal, which are coded.

Signal timing



With the start of the STROBE signal, the above digital signals are read to select an appropriate program. After the program has been selected, the ACK signal turns on. The “ACK” signal is not automatically turned off; it is turned off with a falling edge of the “STROBE” signal. Turn on the RUN signal and execute the program.

MEMO

1. For details on how to register to the program file name and program number and the bit assignments for the digital input signals, refer to “STEA1166_Parameter Manual 4. EXTRNSEL.PAR”. This signal can be used only in the AUTO mode.

2. The STROBE signal should not be input together with the PRG_RST, CYC_RST, STEP_RST or DO_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid. If a file other than the current file is selected, the program is reset to step 1 and the values of variables are reset also.

* Program file name and register of it to program number, and assignment of bits to external digital input signals:

  Bits must be assigned to select executable files for programming using digital or extended input signals. For more information, refer to the “STEA1166_Parameter Manual, 4. EXTRNSEL.PAR”.

[X000] SPECIFY SIGNAL FOR EXTSELECT
FILE-SEL-SIG = (Signal No.), (bit length)
'{Table No.0}
FILE[0].DRIVE = (Save to)[1)]
FILE[0].NAME =("Program file name")
1) 0: SD card 1: Controller

Setting example

```
‘[X000] SPECIFY SIGNAL FOR EXTSELECT
FILE-SEL-SIG = 2,3
```

```
'{Table No.0}
FILE[0].DRIVE = 1
FILE[0].NAME = "PROG000"
FILE[1].DRIVE = 1
FILE[1].NAME = "PROG001"
FILE[2].DRIVE = 1
FILE[2].NAME = "PROG002"
FILE[3].DRIVE = 1
FILE[3].NAME = "PROG003"
```

When set as shown above, if "STROBE" is set to ON, the assigned program is selected based on the status of DIN2 to DIN4. If DIN2 only is ON, PROG001 is selected. If DIN2 and DIN3 are ON, PROG003 is selected.

DIN2	DIN3	DIN4	
ON	OFF	OFF	PROG001 is selected
ON	ON	OFF	PROG003 is selected

MEMO

1. If a parameter was changed, it will not be enabled unless you select "SAVE", and then turn the power off and on again.

5.3.2 PRG-RST (Program reset) Signal

・INPUT pin number 7

Used to reset a currently stopped program to step 1. The value of each variable is also reset to zero (0).

This signal can be used only in the AUTO mode.

MEMO

1. The PRG_RST signal should not be input together with the STROBE, CYC_RST, STEP_RST or DO_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid.

2. This signal cannot be used while AUTORUN (automatic operation mode ON) is set ON.

5.3.3 STEP-RST (Step reset) Signal

・INPUT pin number 26

Used to reset a currently stopped program to step 1. The value of each variable used in the program remains unchanged.

This signal can be used only in the AUTO mode.

MEMO

1. The STEP_RST signal should not be input together with the STROBE, PRG_RST, CYC_RST or DO_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid.

2. This signal cannot be used while AUTORUN (automatic operation mode ON) is set ON.

5.3.4 CYC-RST (Cycle reset) Signal

・INPUT pin number 8

Used to reset a currently stopped program to the step labeled "RCYCLE". The value of each variable used in the program remains unchanged.

This signal can be used only in the AUTO mode.

MEMO

1. The CYC_RST signal should not be input together with the STROBE, PRG_RST, STEP_RST or DO_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid.

2. This signal cannot be used while AUTORUN (automatic operation mode ON) is set ON.

5.3.5 DO-RST (Command reset) Signal

・INPUT pin number 27

This is a signal that resets the digital output signals (DO_1 to DO_64, DO_101 to DO_164, and DO_301 to DO_2264) of the controller from an external device. When reset, all of DO_1 to DO_64, DO_101 to DO_164, and DO_301 to DO_2264 become OFF. However, the output of the system output signal is retained.
This signal can be used only in the AUTO mode.

MEMO

1. The DO_RST signal should not be input together with the STROBE, PRG_RST, CYC_RST or STEP_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid. Neither system output signals nor hand output signals are reset.

2. This signal cannot be used while AUTORUN (automatic operation mode ON) is set ON.

5.3.6 ALM-RST (Alarm reset) Signal

・INPUT pin number 9

Alarms that occurred in a state where the controller start process can be performed from an external device can be cleared.
This signal can be used only in the AUTO mode.

MEMO

1. If an alarm was output at an emergency stop level where the EX_SVON process cannot be performed, the alarm is not cleared by the ALM_RST signal.

5.3.7 RUN (Run) Signal

・INPUT pin number 28

This is used to start the controller program from an external device for enabling automatic operation to be performed. This signal can be used only in the AUTO mode.

Refer to Fig. 6.3 Timing chart for signal timing.

MEMO

1. After servo ON, automatic operation starts with the start of the "RUN" signal.

2. It takes three (3) seconds from the input of "EX_SVON" to the time when the robot is actually ready to work. Thus, it is necessary to program a delay of 3.5 seconds or over between the "SV_RDY" signal ON and the "RUN" signal ON. If the "RUN" signal is input without a delay, it is neglected and the automatic operation is not started.

5.3.8 EX_SVON (External servo ON) Signal

・INPUT pin number 10

This can turn on the servo from an external device. Once the servo power is turned on, it is maintained even after this signal turns off. This signal can be used only in the AUTO mode.

Refer to Fig. 6.3 Timing chart for signal timing.

MEMO

1. This controller sets the interval time (3 seconds) as the time from receiving the "EX_SVON" signal to outputting the servo ready signal. The interval time is set to prevent internal damage to the controller.

2. After inputting the "EX_SVON" signal, ensure a delay of at least 3.5 seconds before inputting the "SVOFF" or "RUN" signal. If the "RUN" signal is input within this period, it will be ignored, and automatic operation will not start.

3. If the "SVOFF" signal (INPUT pin No. 14) is not shorted, the servo-on operation will not start even if the servo-on signal is input.

5.3.9 STOP (Stop) Signal

・INPUT pin number 12

This can stop execution of a controller program from an external device. When this signal becomes open, execution is stopped when the operation command that is currently being executed is completed. When this signal is open, the robot cannot be operated.

This signal can always be used regardless of the master mode.

MEMO

1. When the RUN command is executed after cancel of STOP, the program restarts from the step next to the interrupted step.

2. RUN signal input is ineffective at the input of STOP signal.

3. Unless this signal is used, short-circuit the INPUT-12 signal according to the common wire connection.

5.3.10 CYCLE (Cycle operation mode) Signal

・INPUT pin number 31

This is used to stop the execution program from an external device after one cycle of operation during automatic operation of the controller.

This signal can be used only in the AUTO mode.

5.3.11 LOW_SPD (Low speed command) Signal

・INPUT pin number 13

Used to cause the robot operation speed to low speed from the external equipment.

The robot operates at a low speed (the low speed command is valid) while this signal is open.

The robot operation speed in the low speed mode can be set by means of the parameter. (Initial set value: 25 %)

When this signal is short-circuited, the previously set value (initial set value: 100 %) takes effect again.

This signal can always be used regardless of the master mode.

5.3.12 BREAK (Deceleration and stop) Signal

・INPUT pin number 32

Used to stop the robot motion from the external equipment. The robot slows down and stops at the same time that this signal is open. After the stop, the robot enters a STOP (RETRY) status. Even if this signal is short-circuited again after the stop of robot motion, the robot will not operate. To restart the robot, short-circuit this signal, then execute the RUN command.

If this signal is open, the robot cannot be started.

This signal can always be used regardless of the master mode.

MEMO

1. If the robot is operating, processing of execution is interrupted and the robot slows down and stops.

2. Unless this signal is used, short-circuit the INPUT-32 signal according to the common wire connection.

5.3.13 SVOFF (Servo OFF) Signal

・INPUT pin number 14

Used to turn off the servo from an external device. When this signal is open, the servo is turned off.

This signal can always be used regardless of the master mode.

MEMO

1. While this signal is open, the servo power cannot be turned on in any mode.

2. Unless this signal is used, short-circuit the INPUT-14 signal according to the common wire connection.

5.4 Input Jumpers

Of the system input signal cables, the following signals are used for the safety measures. The robot will not operate unless the following signal lines are jumpered. For this reason, the following INPUT jumper connector1) with the system input signals jumpered, is included as standard. When this connector is used with the TS5000-MS, the internal jumpers are of P24G(-) common type P.

1) Item name: I/O jumper connector

System input signal cables

INPUT-12	(STOP)
INPUT-14	(SVOFF)
INPUT-32	(BREAK)
INPUT-16	(P24V)
INPUT-17	(SYSINCOM)
INPUT-35	(P24G)

Unless the following signals are used as the external signals, jumper the cables also.

INPUT-13	(LOW_SPD)
INPUT-31	(CYCLE)

Jumper of connector

INPUT					
12-16	14-16	32-16	17-35	(18-19)[1)]	(36-37)[1)]

1) This jumper is not used with this controller.
The robot can be operated without any problem with this jumper processing.

MEMO

1. The controller servo power cannot be turned on unless "SVOFF" signal and the emergency stop contact 1 and 2 signals is close.

2. If the CYCLE signal is not connected by a jumper, the controller runs in cycle operation mode.

3. If the LOW_SPD signal is not connected by a jumper, the robot automatic operation speed is low speed.

4. If the STOP signal is not connected by a jumper, robot automatic operation is not performed.

5. For the connectors supplied for the controller, these signals are already connected by a jumper. However, note that the connection is P24G(—) common Type P. Also, to operate the robot without using the system input signal, be sure to connect the supplied connector to the INPUT connector on the controller.

6. When wiring the input signal line on the customer's side, either change the wiring of this connector or obtain the INPUT cable described in 13.1 INPUT Cable.

5.5 Fabricating the INPUT Connector Cable

When fabricating an external I/O signal cable, observe the following matters.

a.) Obtain and use the supplied connector or a connector of the same type for the INPUT connector of the controller. The equivalent of other makers (D-sub connectors) can also be used.

Connector name	Type of connector	Maker
INPUT	XM3A-3721(plug type connector)	OMRON
	XM2S-3711(connector cover)	

b.) Select the cables which meet the following specifications.

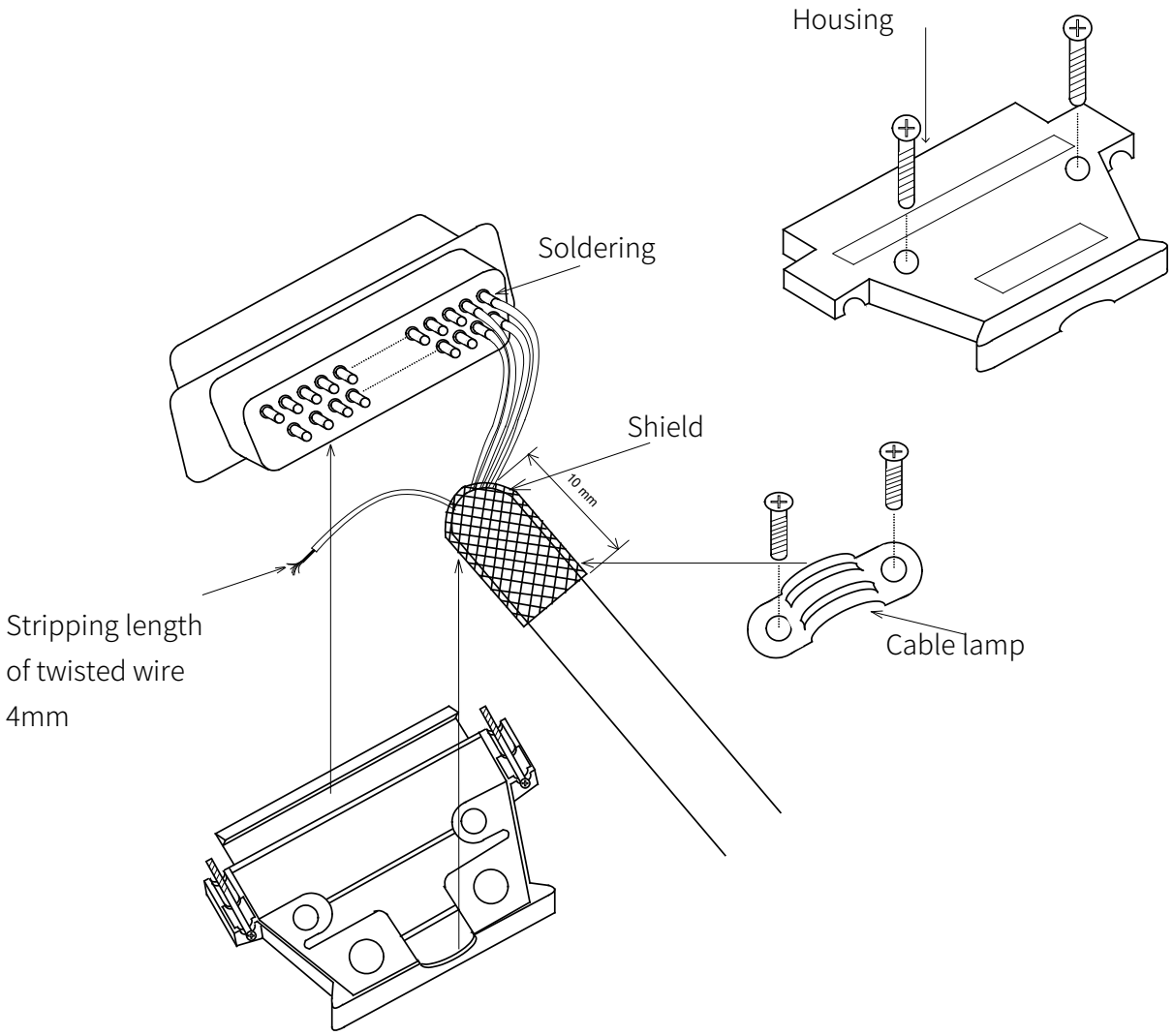
- Core wires : 0.18 mm^2 to 0.32 mm^2 twisted wires
  Withstanding voltage 60 V/1 A or less/75°C
- Cable outer diameter : Max. 13 mm (INPUT), 11 mm (OUTPUT)
- Shield : Batch braided shield
- Cable length : 30 m or less

⚠DANGER

- Never use wires other than those specified. Using a cable that differs from the specifications may cause a fire due to short circuit or overheating of the core wire. Also, the robot may malfunction due to noise intrusion and other factors, and may be struck hard.

c.) The INPUT connector is a solder cup type where the cables are joined by solder.
Fabricate the connection cables for the input/output signal connectors INPUT and OUTPUT as shown in the drawing below.
The shield of the cable using the D-sub connector is connected to the ground of the robot controller by passing through the housing. Completely connect the shield wire to the housing using the cable clamp. Also, it is recommended to attach a protective cover such as thermal contraction tube to the soldered part to prevent a short-circuit.

* A cable where one end is soldered onto the connector is also available as an INPUT option cable. For more information, see 13.1 INPUT Cable, and then contact SHIBAURA MACHINE Customer Service or Sales Department at the headquarters.



5.6 Attaching and Detaching INPUT Connector

Before attaching or detaching the INPUT connector from the controller, be sure to always shut off the main power in the control panel of the customer.

To connect the INPUT connector, as shown in Fig. 5.3 Attaching and Detaching INPUT Connector, fully insert the cable-side connector into the controller-side connector, and then use a screwdriver to tighten the lock screws at both ends of the connector. To disconnect the cable, loosen the lock screws, then pull out the connector.

At this time, DO NOT pull the cable instead of the connector. Otherwise, an excessive force will be exerted on the cable, resulting in damage of the cable.

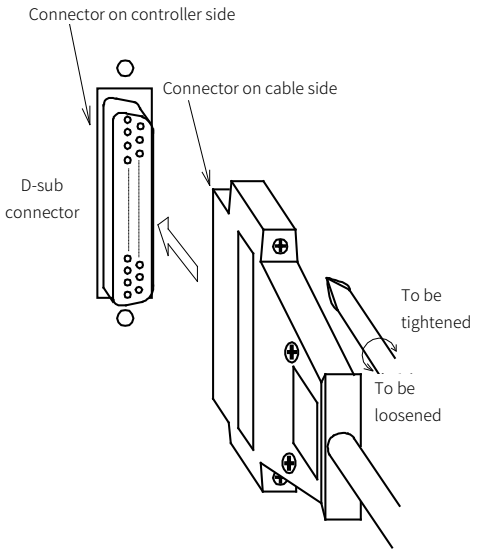


Fig. 5.3 Attaching and Detaching INPUT Connector

6. Output Signal Cable

The cables must be connected by qualified personnel. Also, these processes must be compliant with the laws, regulations, and ordinances of the respective country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

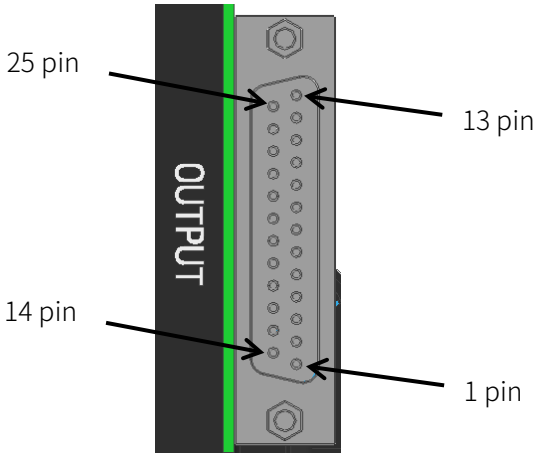
⚠DANGER

- Never use unauthorized cables. Manufacturing and use of unauthorized cables can result in a fire.
- Do not insert connectors or connect to terminals that are not specified. Unspecified wiring causes fires. Also, electric shock may occur due to leakage of electricity, etc.
- Make sure to connect a ground wire. If the ground wire is not connected, there is a risk of electric shock due to leakage of electricity.
- Make sure that the connectors are securely connected. If the mating is loose, static electricity may enter and cause the robot to malfunction and collide.

6.1 Connection of External Output Signal Cable

The external output signal cable is connected using the connectors supplied for the cables [XM3A-2521 (plug type), XM2S-2511 (connector cover)]. A dust cover is attached to the connector at shipping.

Printed board-side connector: XM3B-2522-112*OMRON Corporation



Pin No.	Signal name	Signal description	Pin No.	Signal name	Signal description
1	OUTPUT0	General-purpose output 1	14	OUTPUT1	General-purpose output 2
2	OUTPUT2	General-purpose output 3	15	OUTPUT3	General-purpose output 4
3	OUTPUT4	General-purpose output 5	16	OUTPUT5	General-purpose output 6
4	OUTPUT6	General-purpose output 7	17	OUTPUT7	General-purpose output 8
5	OUTCOM	Output common	18	OUTCOM	Output common
6	ACK	Acknowledge output	19	SV_RDY	Servo ready
7	AUTO	During external mode	20	SYS_RDY	Operation ready
8	AUTORUN	During automatic operation	21	CYC_END	Cycle end
9	LOW_ST	During low-speed mode	22	BT_ALM	Battery alarm
10	SYSOUTCOM	Output common	23	SYSOUTCOM	Output common
11	ALARM	Alarm output	24	EXP24V	External P24V power supply
12	EXP24V	External P24V power supply	25	FUGND	External P24G ground
13	FUGND	External P24G ground			

The output signal cable is connected using the supplied connectors [XM3A-2521 (plug type), XM2S-2511 (connector cover)].

An option is available where one side of the cable is soldered onto the connector. For details, please contact SHIBAURA MACHINE Customer Service or the Headquarters Sales Department. For details about cables, refer to 13.2 OUTPUT Cable.

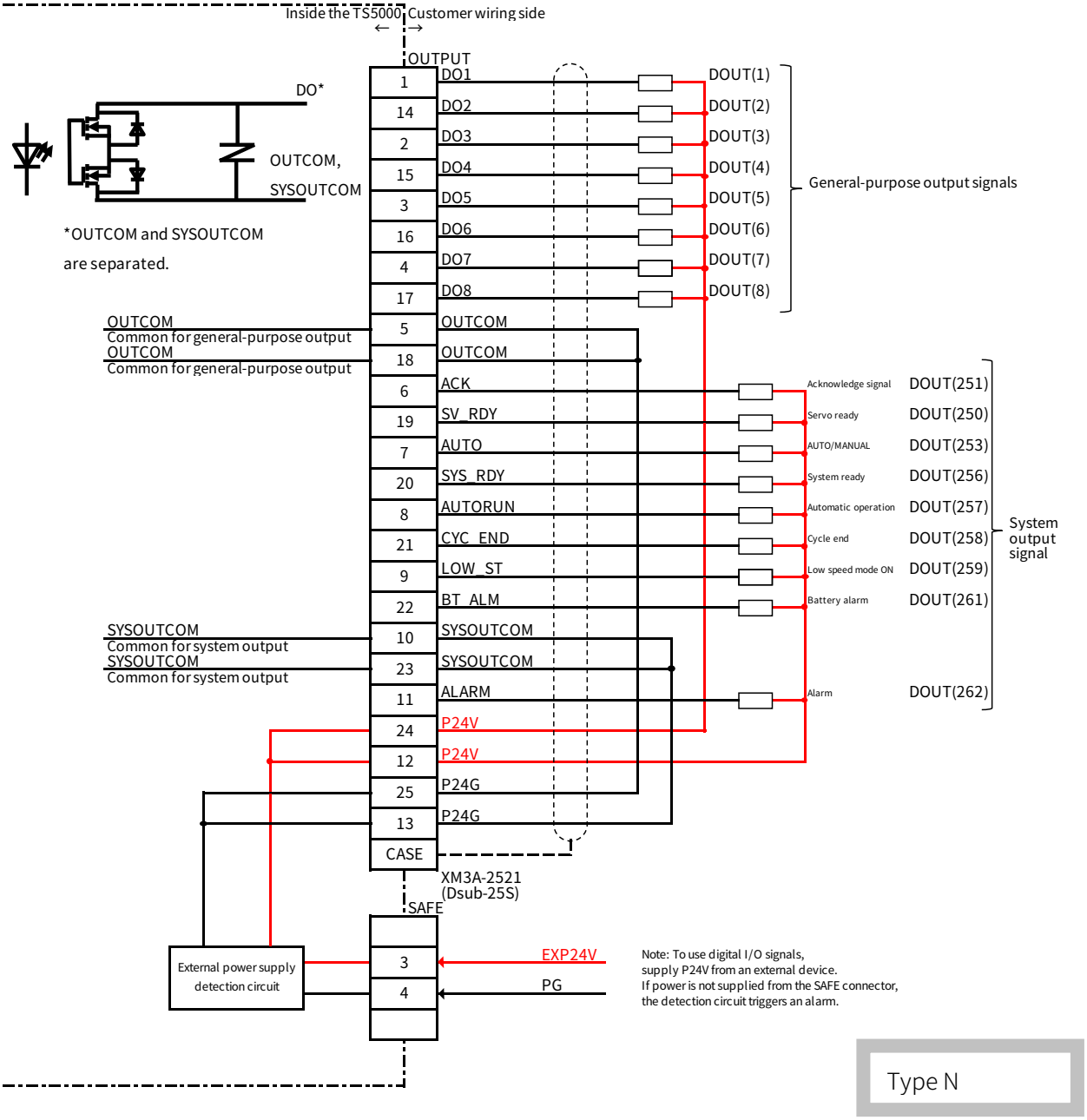


Fig. 6.1 Connecting the Output Signal Cables (Type N)

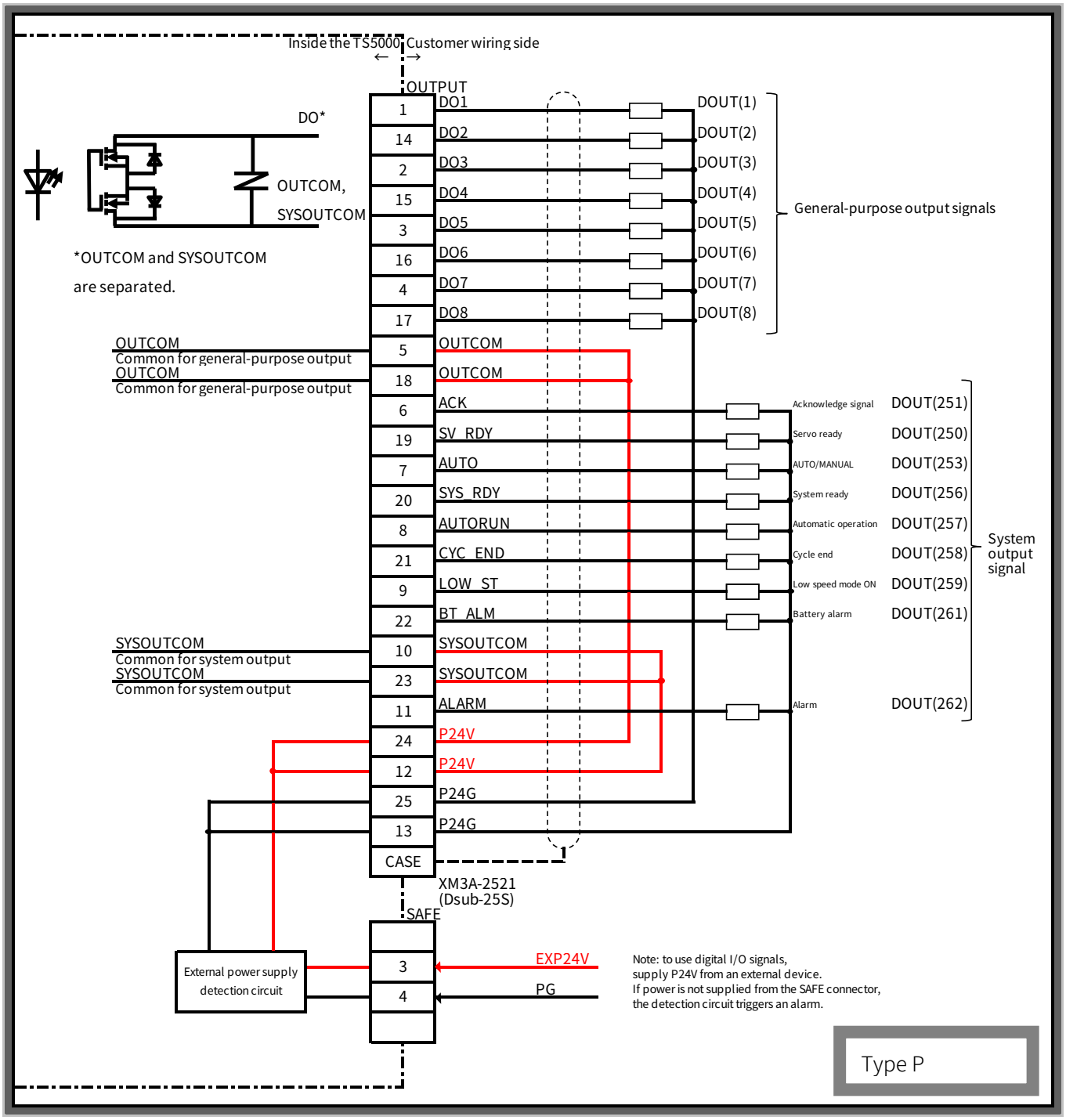


Fig. 6.2 Connecting the Output Signal Cables (Type P)

MEMO

1. All output signals are 17 digital output signals and are PhotoMOS outputs. For more information on the functions, circuits used, and other details for each signal, refer to the following pages. Also, external output control is possible using TCmini (internal PLC). For details, refer to the "STEA1168_TCmini (Internal PLC) Manual".

6.2 General-purpose output signals

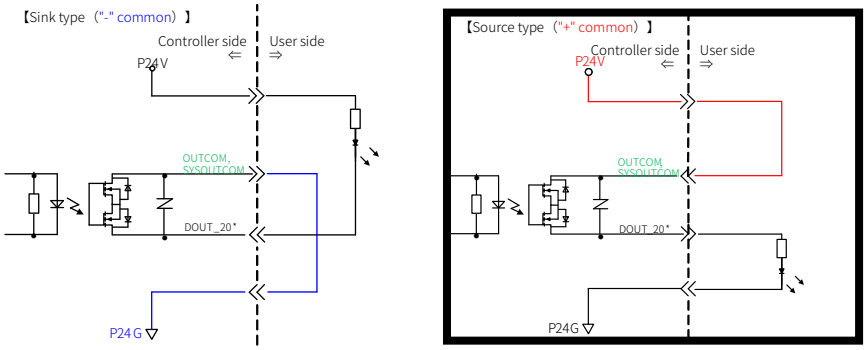
・OUTPUT pin number 1 to 4, 14 to 17

DOUT1 to DOUT8 of the COM line are OUTCOM.

ON/OFF of signals DO_1 ~ DO_8 and pulse output can be performed by the robot program (BCDOUT command and PULOUT command).

Output type: Photo MOS output (bi-direction)

Output circuit structure



NOTICE

- Be sure to always use within the rated output current. If a current exceeding the rated output is supplied, the output elements, printed boards, or other components could be damaged, resulting in a fire.

6.3 System Output Signal

・OUTPUT pin number 6 to 9, 11, 19 to 22

The system output signals of the COM line are SYSOUTCOM.

The system output signal is a 9-point signal that notifies external devices of the operation ready signal “SYS_RDY” and servo ready signal “SV_RDY”. A dedicated output terminal is set for each function. The system output signal’s configuration is the same as the general-purpose output signal’s circuit configuration, so refer to 6.2 General-purpose output signals.

Table 6.1 List of the system output signals

Name	Signal name in DOUT command	Function	Signal judgment
ACK	251	Serves as a response signal to input signals STROBE, PRG_RST, STEP_RST, CYC_RST and DO_RST.	ON : HIGH OFF: LOW
SV_RDY	250	This signal is output when the servo power is turned on and the robot is ready to start.	ON : HIGH OFF: LOW
AUTO	253	Master mode status output AUTO mode when ON, MANUAL mode when OFF	ON : HIGH OFF: LOW
SYS_RDY	256	System ready signal	ON : HIGH OFF: LOW
AUTORUN	257	This signal is output while the program is executed in automatic operation.	ON : HIGH OFF: LOW
CYC_END	258	This is the output signal for verifying that the program has stopped by the CYCLE signal input.	ON : HIGH OFF: LOW
LOW_ST	259	This signal is output while the robot is operating at a low speed by the input of low speed command.	ON : HIGH OFF: LOW
BT_ALM	261	Battery voltage level error output for backup power supply.	ON : HIGH OFF: LOW
ALARM	262	Controller fault output.	ON : HIGH OFF: LOW

The figure below shows the timing chart for TS5000. Once the controller power is turned on and the system boots up, the “SYS_RDY” signal, indicating the system is ready for operation, turns ON. Once the “EX_SVON” input is received, the servo turns on. Then, 3 seconds after the input, the “SV_RDY” signal is output. Once the “SVOFF” signal is turned OFF, the controller turns the servo off, and the “SV_RDY” output is turned OFF.

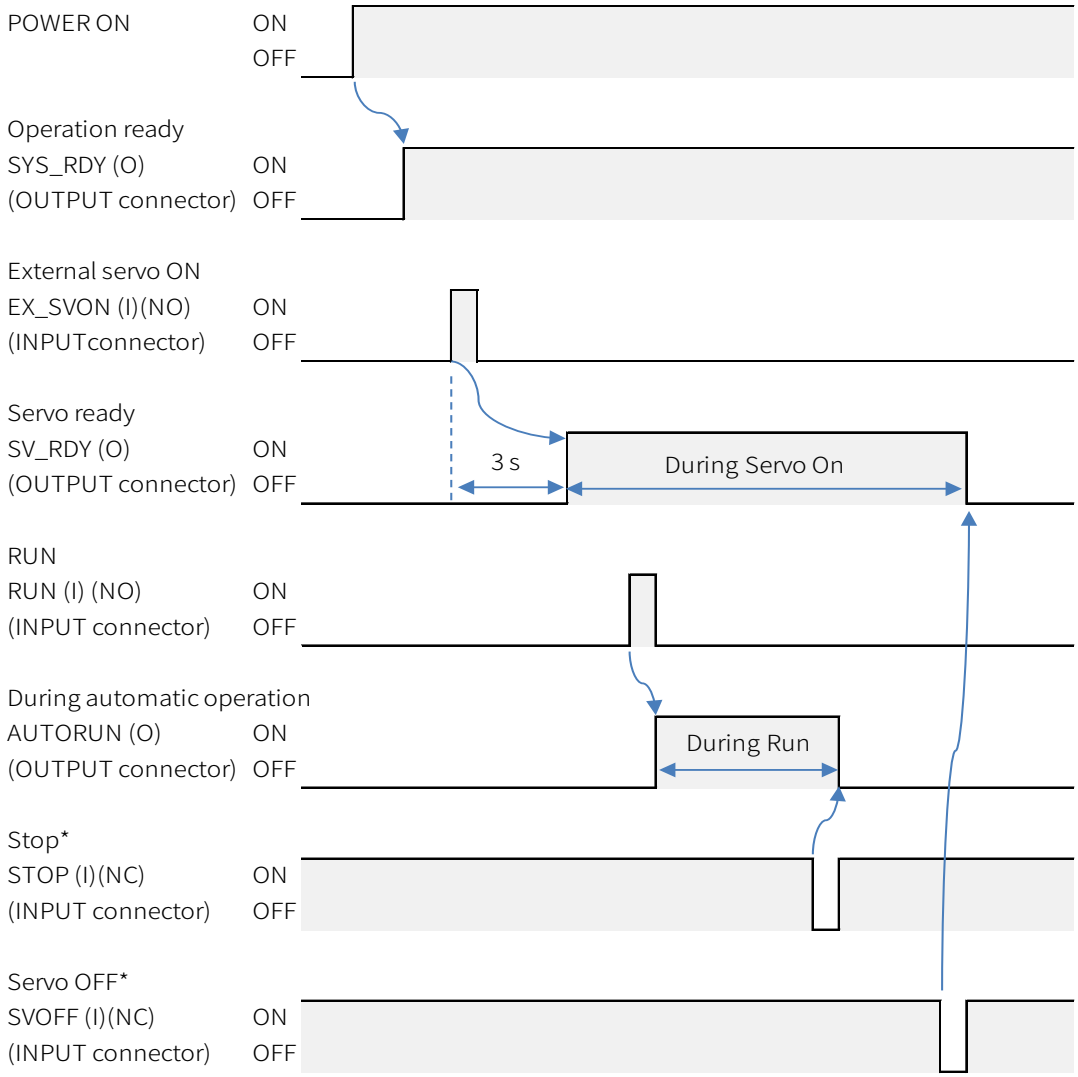


Fig. 6.3 Timing chart

6.3.1 ACK (Acknowledge output) signal

・OUTPUT pin number 6

This signal is a response signal to the input of either “STROBE”, “PRG_RST”, “STEP_RST”, “CYC_RST” or “DO_RST” signal. When either of the previous signals is input, the “ACK” signal indicates that processing has been completed. This signal can always be used regardless of the master mode.

MEMO

1. When each input signal (“STROBE”, “PRG_RST”, “STEP_RST”, “CYC_RST”, “DO_RST”) is input at the same time, only the first input signal is processed and the “ACK” signal is output.

6.3.2 SV_RDY (Servo ready) signal

・OUTPUT pin number 19

This signal indicates that the controller servo power is turned on, and the robot is in a state where operation is possible. This signal can always be used regardless of the master mode. It also outputs the signal at the same timing as the servo ON status contact (Pin No. 13, 14 SVST) of the SAFE connector. Refer to Fig. 6.3 Timing chart for signal timing.

MEMO

1. Program so that the “RUN” signal can be turned on only after the “SV_RDY” signal turns on.

6.3.3 AUTO (During AUTO mode) signal

・OUTPUT pin number 7

This signal is set to ON when the controller is in AUTO mode.
When this signal is OFF, the controller is in MANUAL mode.

MEMO

1. When master mode is set to AUTO mode, all system input signals can be used.

6.3.4 SYS_RDY (System ready) signal

・OUTPUT pin number 20

After the power supply circuit breaker is turned on so that power is supplied, this signal turns on when the controller was started normally. With this signal, it is possible to confirm that the robot can be operated.
This signal can always be used regardless of the master mode.
Refer to Fig. 6.3 Timing chart for signal timing.

MEMO

1. When the operation mode is set to AUTO mode, all system input signals can be used.

6.3.5 AUTORUN (Auto mode ON) signal

・OUTPUT pin number 8

This signal turns on when the robot is operating in the automatic operation mode.
When the RUN signal is made valid after the SYS_RDY signal is turned on, automatic operation starts with this signal output.
This signal remains on as long as the robot is operating in the automatic operation mode.
This signal turns on when in AUTO mode.
Refer to Fig. 6.3 Timing chart for signal timing.

MEMO

1. This signal does not turn on when in MANUAL mode.

6.3.6 CYC_END (Cycle end) signal

・OUTPUT pin number 21

This signal turns on after the stop of 1-cycle automatic operation only when the CYCLE signal is made valid in the program execution (RUN) mode and an automatic operation is executed.
This signal can always be used regardless of the master mode.

MEMO

1. This signal turns off when the RUN mode takes effect.

2. This signal turns on also when the cycle operation has been stopped by the STOP, BREAK or ALARM signal.

6.3.7 LOW_ST (Low speed mode ON) signal

・OUTPUT pin number 9

This signal turns on while the robot is operating in the low speed mode by the input of system input signal LOW_SPD.
This signal can always be used regardless of the master mode.

MEMO

1. This signal turns off when the program execution "RUN" becomes possible.

2. This signal is also turned on when stopped by either "STOP", "BREAK" or "ALARM".

6.3.8 BT_ALM (Battery alarm) signal

・OUTPUT pin number 22

This signal turns on if a battery alarm has occurred in the robot or robot controller.
The battery alarm is triggered by detecting the internal battery level for all axis encoders (from 1st axis to 4th axis) and the motherboard (X8KM).
This signal can always be used regardless of the master mode.

MEMO

1. When a battery alarm occurs, the location of the battery where the voltage level has dropped is identified by the alarm code displayed in the error message, and so replace and inspect the battery as soon as possible by referring to "STEA1161_Controller Manual and STEA1171_Robot Manual".

6.3.9 ALARM (Alarm) signal

・OUTPUT pin number 11

This turns on when a level 2, 4, or 8 alarm has occurred in the controller or robot body.
This signal is kept ON during error detection and turns off after the error has been cleared.
For details on the faults, refer to “STEA1165_Alarm Manual”.
This signal can be used regardless of the master mode.

6.4 Fabricating the OUTPUT Connector Cable

When fabricating an external I/O signal cable, observe the following matters.

a.) Use the included connector or a connector of the same type for the controller's OUTPUT connection connector. The equivalent of other makers (D-sub connectors) can also be used.

Connector name	Type of connector	Maker
OUTPUT	XM3A-2521(plug type connector) XM2S-2511(connector cover)	OMRON

b.) Select the cables which meet the following specifications.

- Core wires : 0.18 mm^2 to 0.32 mm^2 twisted wires
  Withstanding voltage 60 V/1 A or less/75℃
- Cable outer diameter : Max. 13 mm (INPUT), 11 mm (OUTPUT)
- Shield : Batch braided shield
- Cable length : 30 m or less

⚠DANGER

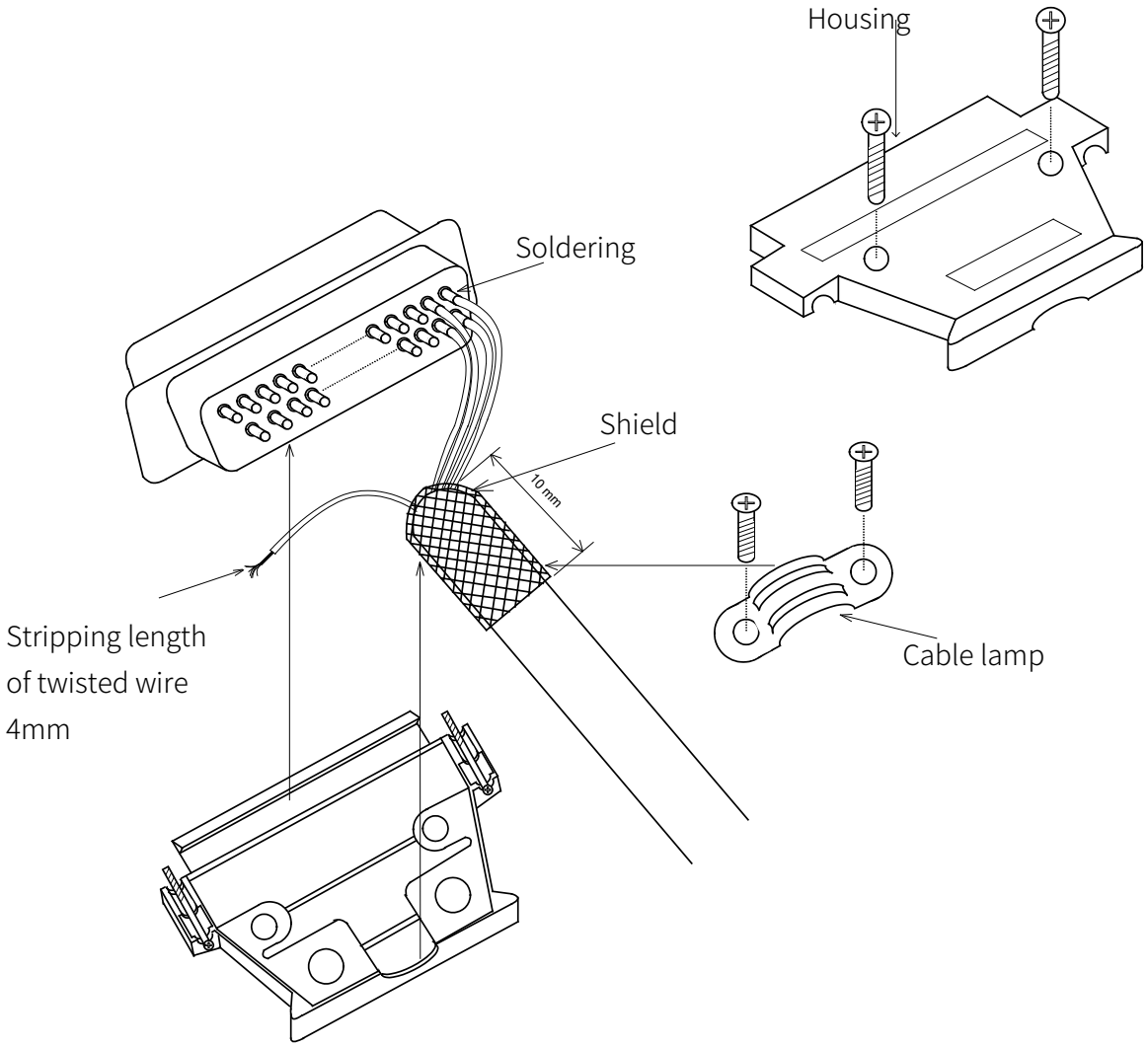
- Never use wires other than those specified. Using a cable that differs from the specifications may cause a fire due to short circuit or overheating of the core wire. Also, the robot may malfunction due to noise intrusion and other factors, and may be struck hard.

c.) The OUTPUT connector is a solder cup type where the wire is joined by solder.

Fabricate the connection cables for the input/output signal connectors INPUT and OUTPUT as shown in the drawing below.

The shield of the cable using the D-sub connector is connected to the ground of the robot controller by passing through the housing. Completely connect the shield wire to the housing using the cable clamp. Also, it is recommended to attach a protective cover such as thermal contraction tube to the soldered part to prevent a short-circuit.

* A cable where one end is soldered onto the connector is also available as an OUTPUT option cable. (We can advise of the cable length.) Contact our after-sale service agent in your territory.



6.5 Attaching and Detaching OUTPUT Connector

Before attaching or detaching the OUTPUT connector from the controller, be sure to always shut off the main power in the control panel of the customer.

To connect the OUTPUT connector, as shown in Fig. 6.4 Attaching and Detaching OUTPUT Connector, fully insert the cable-side connector into the controller-side connector, and then use a screwdriver to tighten the lock screws at both ends of the connector. To remove the connector, do the reverse of the connection; loosen the locking screw and then pull out the connector. If the wire section is pulled out without holding the connector, excessive force will be applied to the cable, causing cable failure.

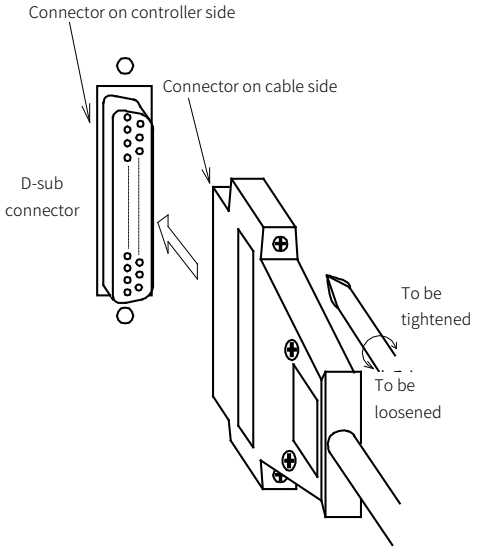


Fig. 6.4 Attaching and Detaching OUTPUT Connector

6.6 Definition of Colors for Operation Buttons and Indicator Lamps

If switches or other devices for operating the controller will be located outside the controller by the customer, please use the following colors for the operation buttons.

- Operation buttons
  Red - Emergency
  Yellow - Trouble
  Green - Normal
  Blue - Required action (e.g., reset)

- Indication of operation buttons

Start	Stop	Pushbutton which alternately starts and stops	Button which allows operation only when held down, and stops it when released
			

- It is requested that the colors of indicator lamps be defined as follows:
  Red - Emergency
  Yellow - Trouble
  Green - Normal
  Blue - Required action (e.g., reset)

7. Teach Pendant Cable

The cables must be connected by qualified personnel. Also, these processes must be compliant with the laws, regulations, and ordinances of the respective country.

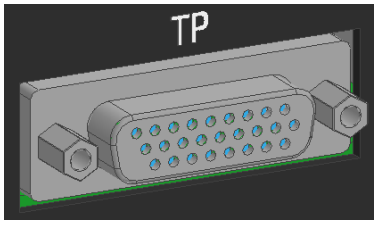
Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

7.1 Overview of TP Cables

The TS5000-MS can be connected to the TP5000 teach pendant for operation. Also, a special connector can be purchased as an option for enabling connection of the TP1000. For details on how to operate the teach pendant, refer to “STEA1162_TP5000 Operation Manual” or “STEA1169_TP1000 Operation Manual”.
A connector with a jumpered emergency stop signal is included in the shipment. When the teach pendant is not connected, connect the supplied dummy plug.
For information on attaching and detaching the INPUT connector, see 5.6 Attaching and Detaching INPUT Connector.

Printed board-side connector: 09 56 252 7613* Harting K.K.



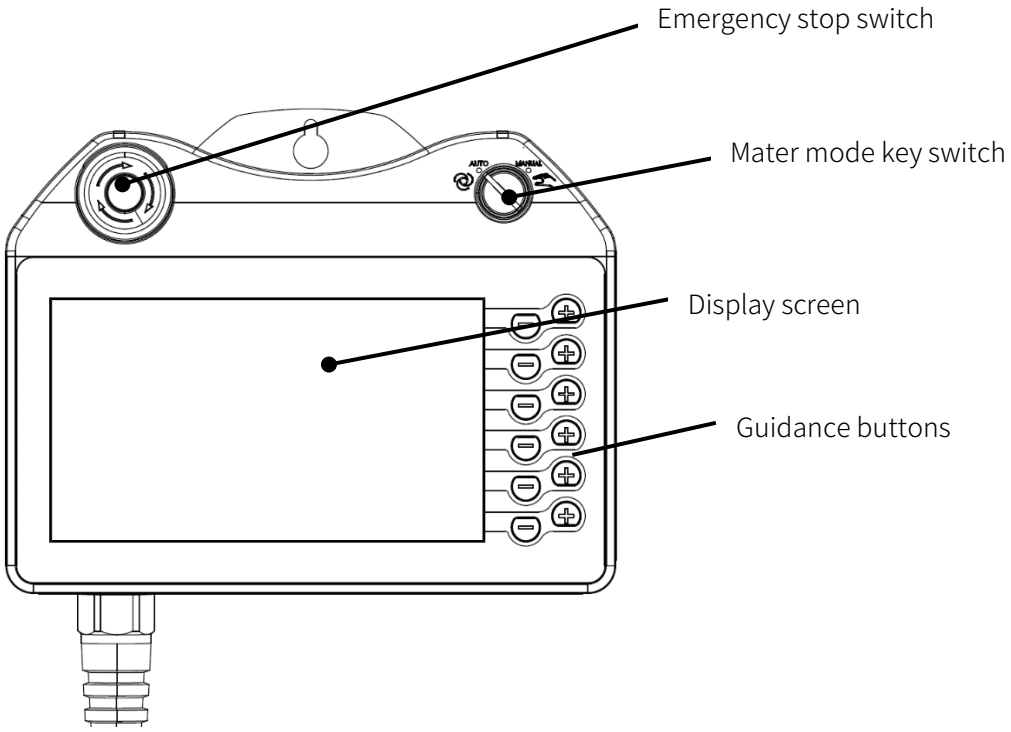
Pin No.	Signal name	Signal description	Pin No.	Signal name	Signal description
1	P24V	Internal 24V output	14	Reserve	Reserve
2	P24V	Internal 24V output	15	Reserve	Reserve
3	TP_MOD1B	Mode 1b contact signal	16	Reserve	Reserve
4	TP_MOD1C	Mode 1 common signal	17	RD+	Ethernet receive +
5	Reserve	Reserve	18	RD-	Ethernet receive -
6	Reserve	Reserve	19	EXT_EMG1	TP emergency stop signal 1
7	GND	P5VG ground	20	OU_EMG1	TP emergency stop signal 1
8	TD+	Ethernet send +	21	EXT_EMG2	TP emergency stop signal 2
9	TD-	Ethernet send -	22	OU_EMG2	TP emergency stop signal 2
10	PG	Internal P24G ground	23	TP_ENA1A	TP enable signal 1
11	PG	Internal P24G ground	24	TP_ENA1C	TP enable 1 common
12	TP_MOD2B	Mode 2b contact signal	25	TP_ENA2A	TP enable signal 2
13	TP_MOD2C	Mode 2 common signal	26	TP_ENA2C	TP enable 2 common

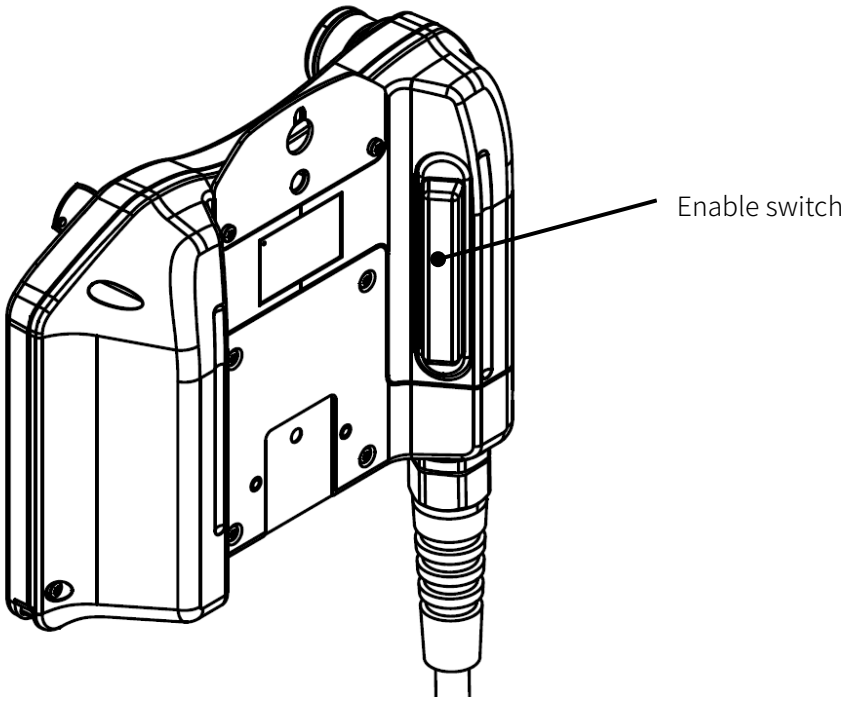
MEMO

1. Note that the servo cannot be set to ON unless a teach pendant or the dummy plug is connected to the TP connector.

7.2 TP5000

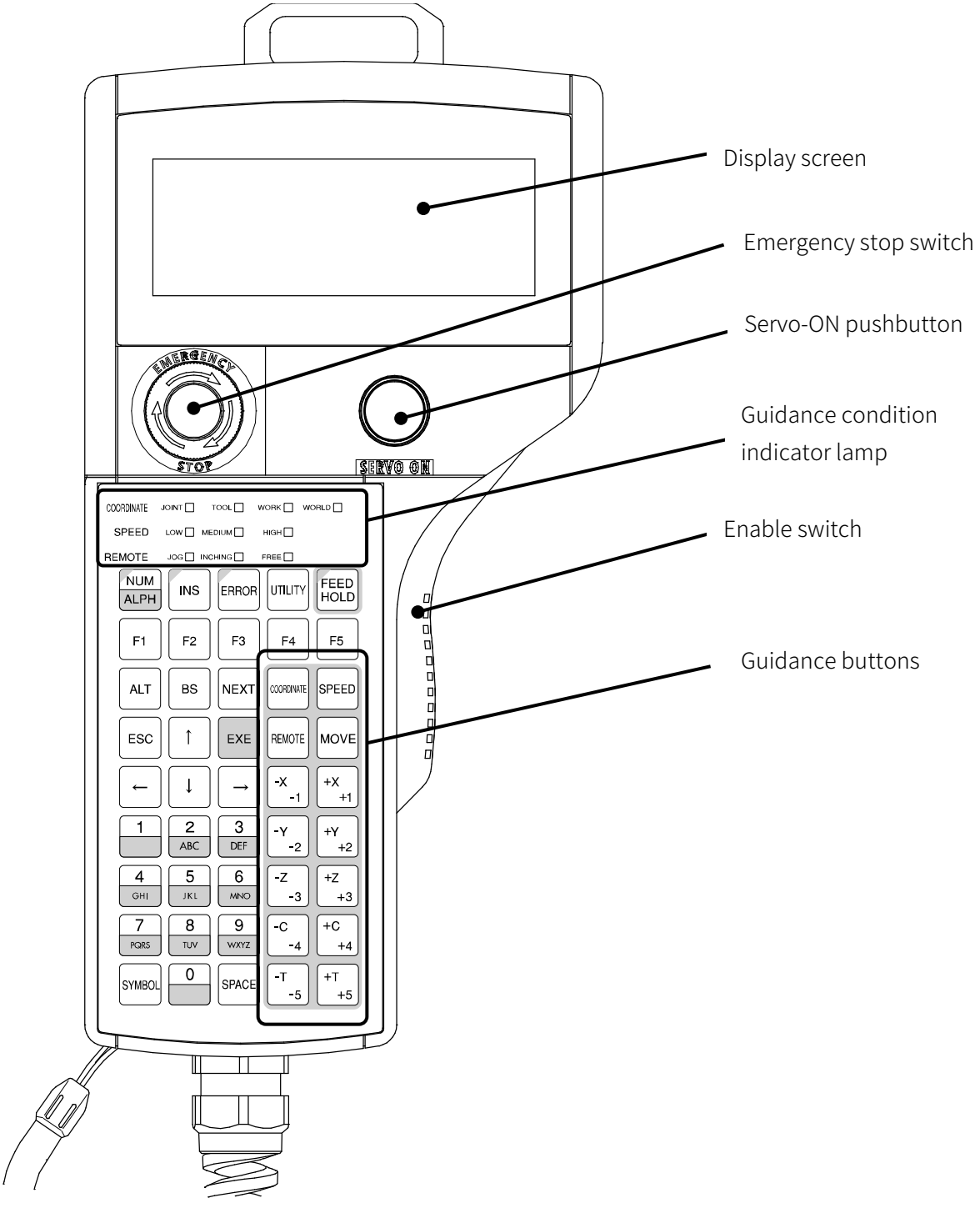
7.2.1 Appearance and Names of Parts





7.3 TP1000

7.3.1 Appearance and Names of Parts

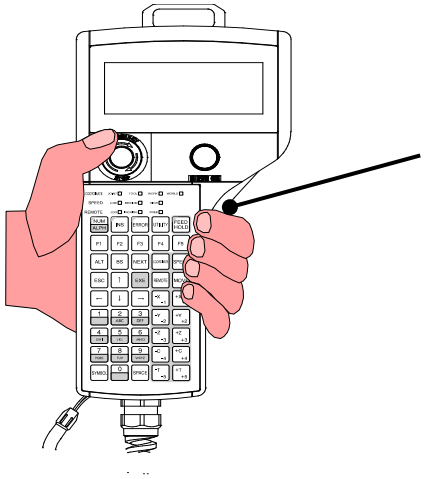


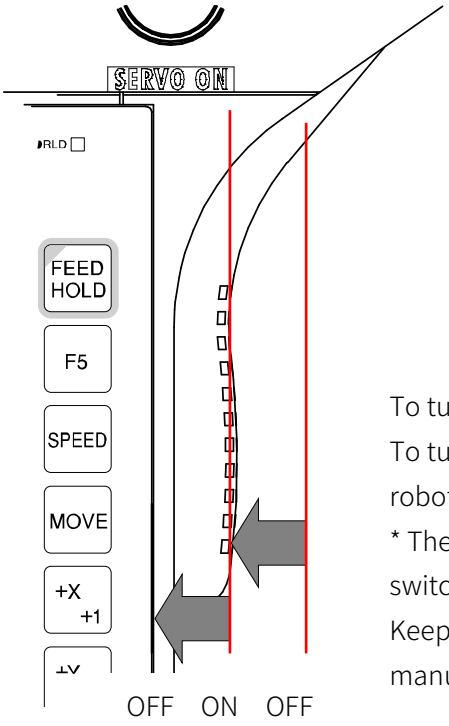
MEMO

1. To use a TP1000, a special conversion cable is required.

7.3.2 Usage Procedure

The TP1000 (Teach Pendant) is equipped with a 3-position Enable switch. Use the Enable switch to turn on the servo in teach mode or guide the robot in teach mode.





To turn on the Enable switch, gently hold the switch.
To turn off the Enable switch, grip on the switch. (The robot stops.)
* The robot cannot be guided manually when the Enable switch is OFF.
Keep the Enable switch ON when guiding the robot manually.

7.4 TP Jumper Connectors

The controller includes connectors for TP jumpers. When the teach pendant is being used, inserting this connector establishes a jumper connection with the emergency stop line so that automatic operation can be performed from external signals.

TP Jumper Connectors

TP			
19-20	21-22	23-24	25-26

8. SD Card Slot

In the standard configuration, the TS5000-MS has one SD slot.

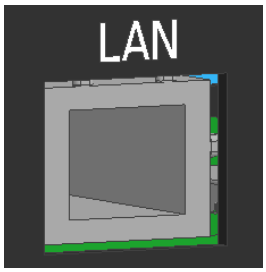


Various commercially available SD cards can be used to transfer or save parameters and backup programs.

> *MEMO*
>
> *1. Use an SD card formatted with FAT16 or FAT32.*

9. Ethernet Connector

In the standard configuration, the TS5000-MS has one Ethernet connector.



This function uses TCP/IP protocol to enable communication with computers or visual devices on the Ethernet within the same local area. Also, the program assistance software TSAssist can be used to perform program creation and editing, assist in debug startup, and perform maintenance.
The robot controller uses an automatic recognition function to enable usage of either a straight-through cable or crossover cable.

For details on the Ethernet communication function, refer to “STEA1164_Communication Manual”.

Table 9.1 Communication specifications

Item		Specifications Description				
Interface		1000BASE-X				
Baud rate		1000 Mbps				
Function		Communication mode	Setting role	No. of available ports	No. of simultaneous connectable clients	Port No.
	Port without any specified communication protocol	TCP / UDP	Server / Client (selectable)	8	1[1)]	10000 to 65535 (selectable)
	Dedicated communication port	TCP / UDP	Server / Client (selectable)	2	1[1)]	10000 to 65535 (selectable)
	Control communication port	TCP	Server (Fixed)	1	1	2000 (Fixed)
	Monitor communication dedicated port	TCP	Server (Fixed)	1	2[2)]	2001 (Fixed)
	Periodic monitor communication port	TCP	Server (Fixed)	1	2[2)]	2002 (Fixed)
	IP dump port 1	TCP	Server (Fixed)	1	1	2003 (Fixed)
	IP dump port 2	TCP	Server (Fixed)	1	1	2004 (Fixed)

1) Ports that were set to the Server function can connect to one client only. Connections from multiple clients are not possible.
2) Up to two clients can be connected simultaneously to the same server port of the controller.

10. Expansion I/O Slot (Option)

By inserting an I/O module (model: TS5K-OPIO) into the expansion I/O slot, you can add 17-21 new general-purpose I/O entries. To consult with us about I/O extension, please contact SHIBAURA MACHINE Customer Service or the Headquarters Sales Department.

10.1 Connecting the Extension Input Signal Cable (EXP-IN)

The expansion input signal cable is connected using the connectors [XM3A-3721 (plug type), XM2S-3711 (connector cover)]. This connector can use the same connector as the input signal cable (INPUT). Also, this connector is not included, and so it must be purchased separately.

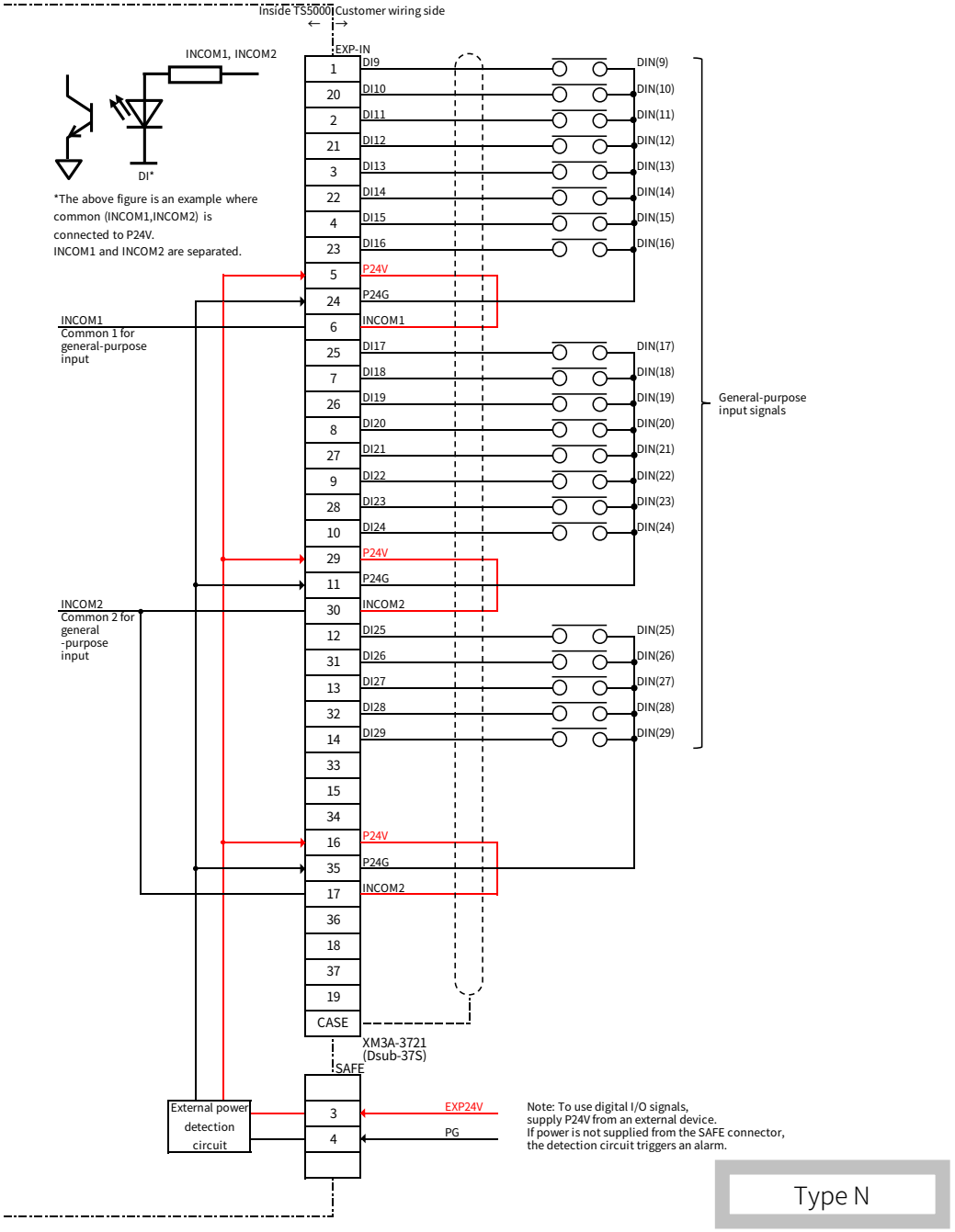


Fig. 10.1 Connecting the Input Signal Cable (Type N)

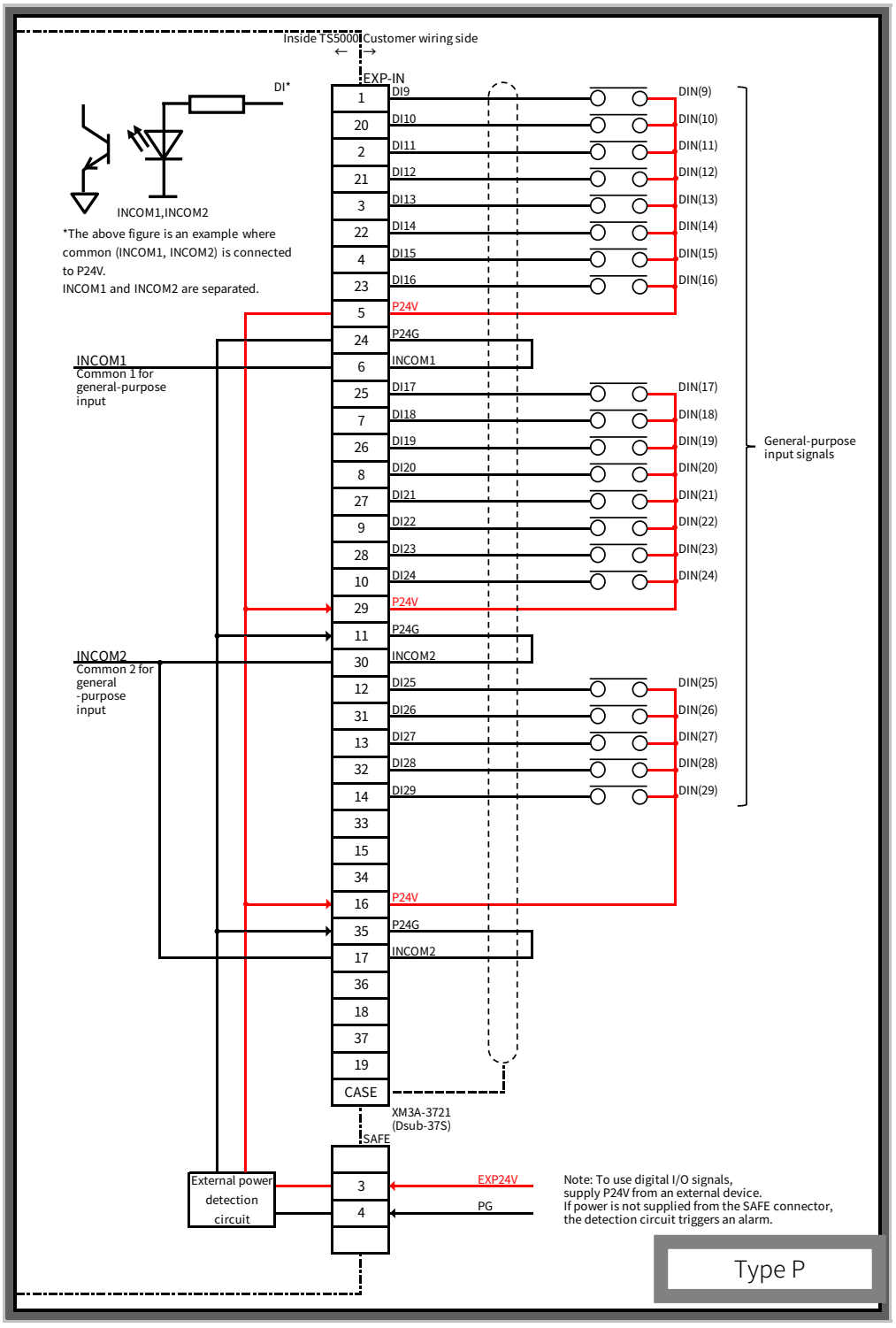


Fig. 10.2 Connecting the Input Signal Cable (Type P)

MEMO

1. The input signals are non-voltage contact input or open collector input, and all are a-contact input. For more information on the functions, circuits used, common line changeover, and other details for each signal, refer to the following pages.

10.2 Expansion input signal DIN9 to DIN29

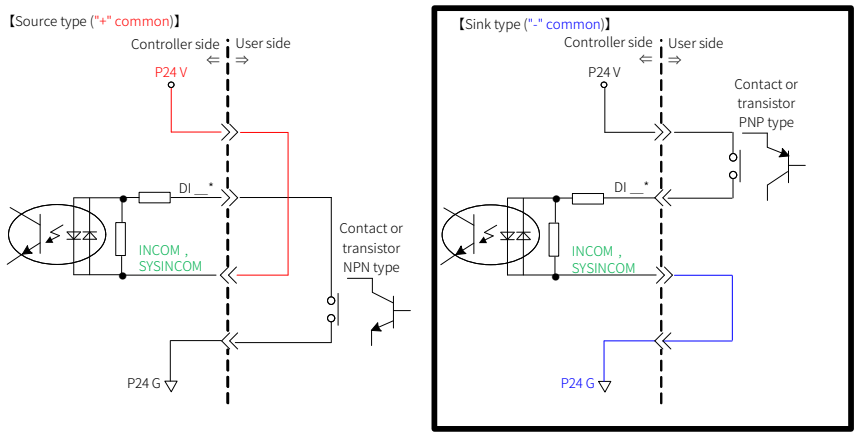
・INPUT pin number 1 to 4, 7 to 10, 12 to 14, 20 to 23, 25 to 28, 31, 32

The common line for DI_9 to DI_16 is INCOM1, and the common line for DI_17 to DI_29 is INCOM2.
The robot program (DIN command) can be used to determine the signal status for DI_9 to DI_29, and the program processes can be branched. Also, interrupt processes can also be performed by changes in the signal that monitored the signals during robot operation.

Specifications of non-voltage contact and transistor

Non-voltage contact specifications	Transistor specifications
Contact rating 24 VDC, 10 mA or over Circuit current Approx. 7 mA Minimum current 24 VDC, 1 mA Contact impedance 100 Ω or less	Withstand voltage between collector and emitter 30 V or over Circuit current: Approx. 7 mA Current between collector and emitter 10 mA or over Leakage current between collector and emitter 100 μA or less

Refer to the following diagram for details on the connection method. (example of used circuit)



MEMO

1. When the pulse type input signals are used, the pulse width should be 100 ms or over.

10.3 Connecting the Expansion Output Signal Cable (EXP-OUT)

The expansion output signal cable is connected using the supplied connectors [XM3A-2521 (plug type), XM2S-2511 (connector cover)]. This connector can use the same connector as the output signal cable (OUTPUT). Also, this connector is not included, and so it must be purchased separately.

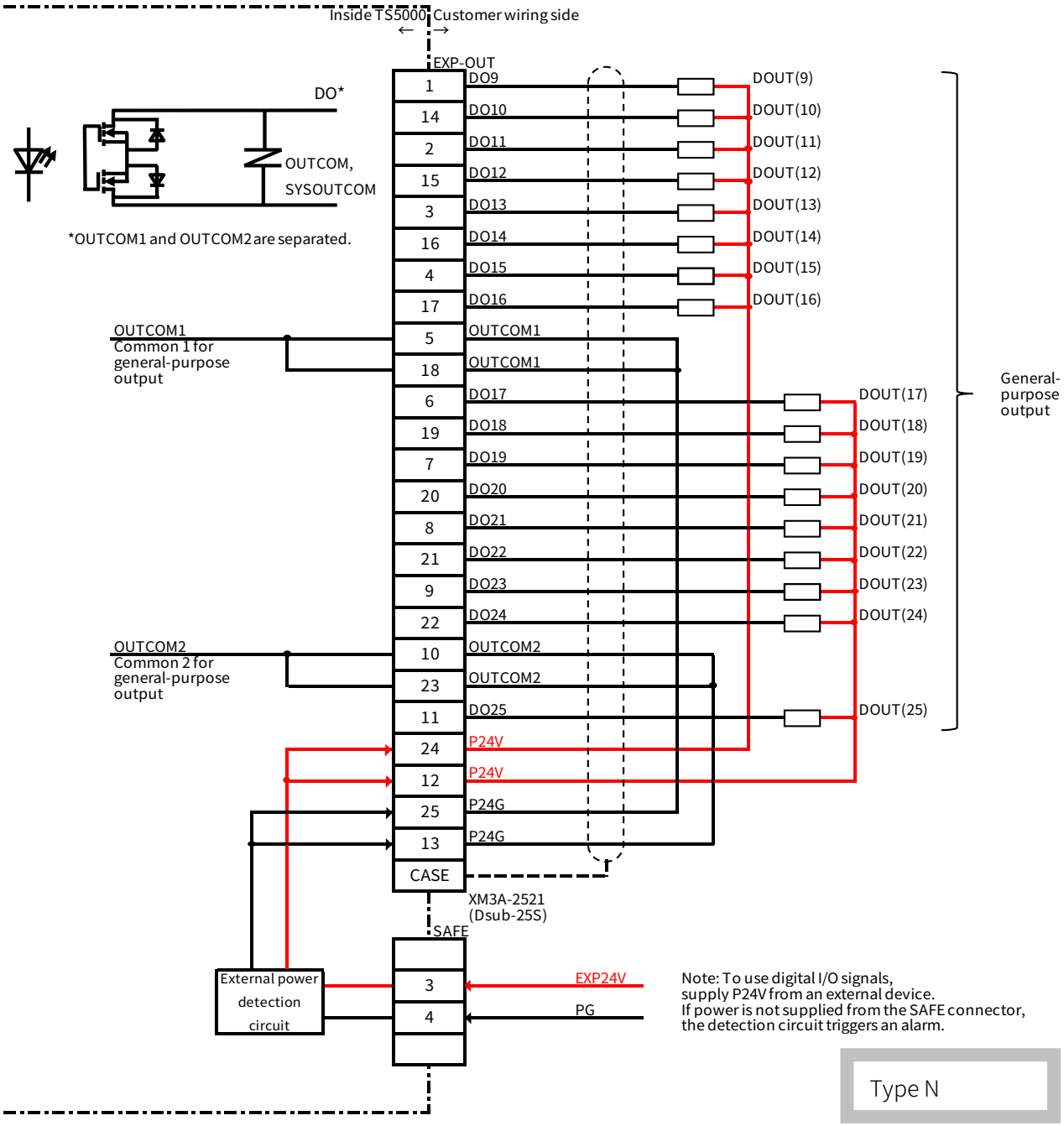


Fig. 10.3 Connecting the Output Signal Cables (Type N)

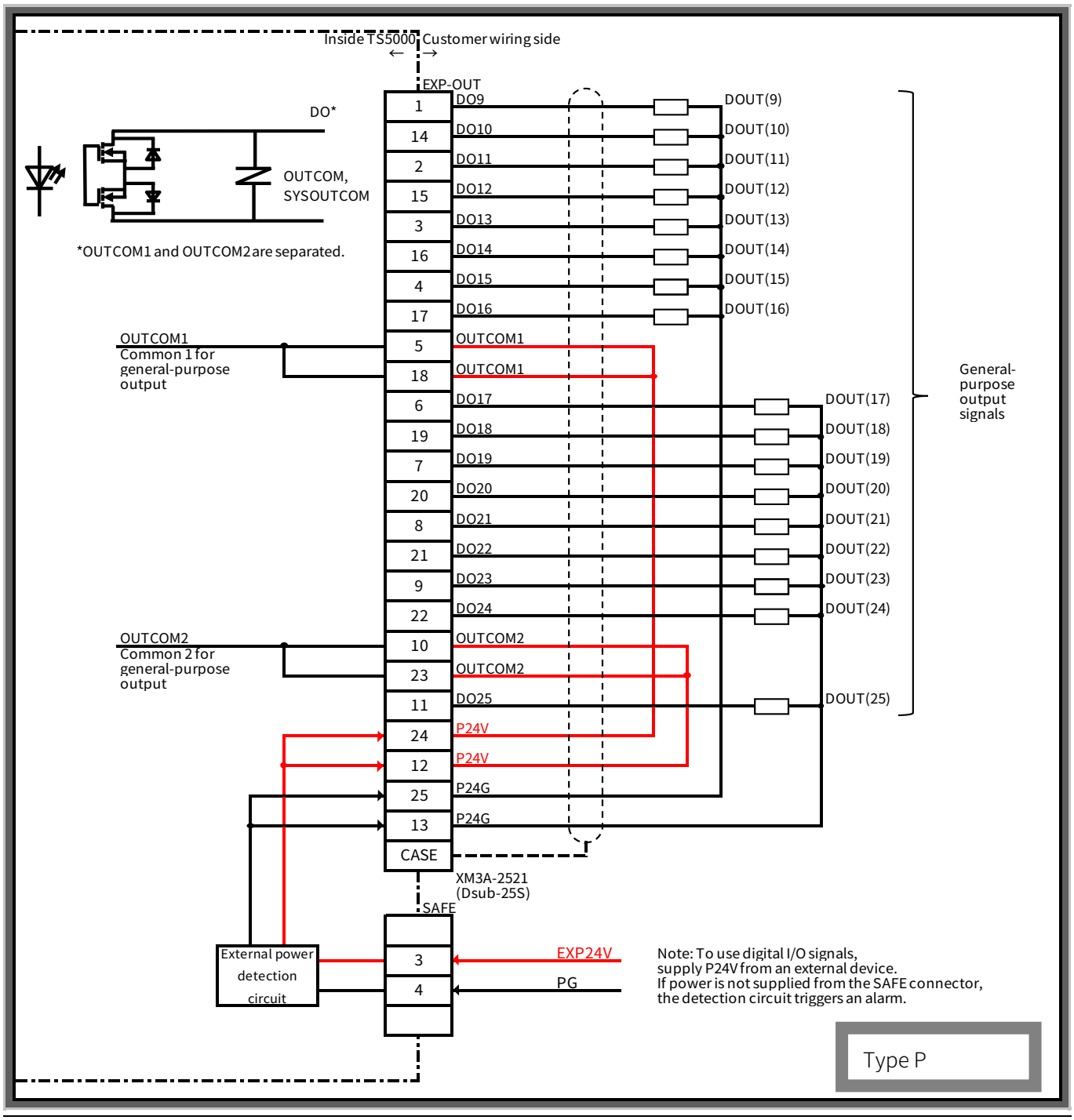


Fig. 10.4 Connecting the Output Signal Cables (Type P)

MEMO

1. All output signals are 17 digital output signals and are PhotoMOS outputs. For more information on the functions, circuits used, and other details for each signal, refer to the following pages. Also, external output control is possible using TCmini (internal PLC). For details, refer to the "STEA1168_TCmini (Internal PLC) Manual".

10.4 Expansion output signal DOUT9 to DOUT25

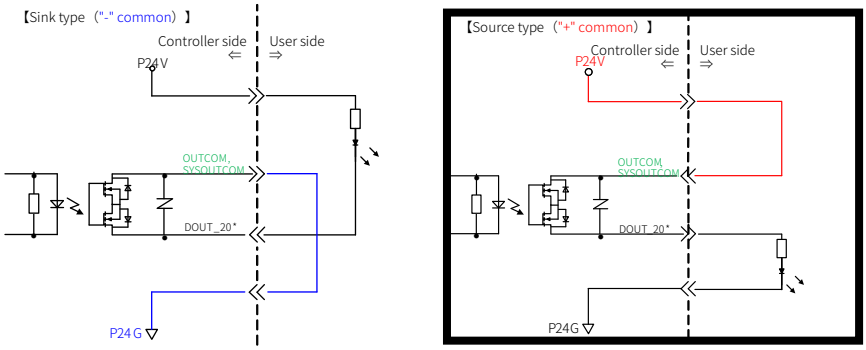
・OUTPUT pin number 1 to 4, 6 to 9, 11, 14 to 17, 19 to 22

DOUT9~DOUT16 of the COM line are OUTCOM1、DOUT17~DOUT25 are OUTCOM2.

ON/OFF of signals DOUT9 to DOUT25 and pulse output can be performed by the robot program (BCDOUT command and PULOUT command).

Output type: Photo MOS output (bi-direction)

Output circuit structure



NOTICE

- Be sure to always use within the rated output current. If a current exceeding the rated output is supplied, the output elements, printed boards, or other components could be damaged, resulting in a fire.

11. TRIG Slot (Option)

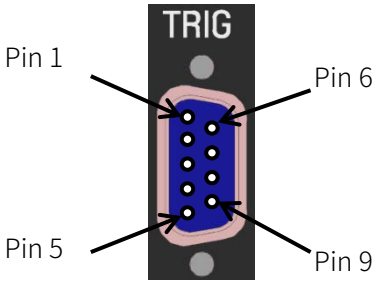
The TRIG slot is for extension of the conveyor synchronization trigger input. A signal cable with 4-point input for trigger input function can be added as an option. When used as a conveyor synchronization start trigger, use 2 points of the high-speed input function 4-point inputs. Connect to the high-speed signal input number: DIN33 for the CH1 conveyor and to the high-speed signal input number: DIN34 for the CH2 conveyor. In addition, the "position data latch function" and "conveyor synchronization function" that use four high-speed signal inputs cannot be used at the same time.

The trigger input signal cable is connected using the optionally supplied connector [XM3A-0921 (plug type) and XM2S-0911 (connector cover)]. Since the trigger input signal configuration is the same as that for the circuit configuration of general-purpose output signal, refer to 5.2 General-purpose input signals DIN1 to DIN8.

For details on vision + conveyor synchronization, refer to “STEA1183_Vision Sensor + Conveyor Synchronization Function Manual”.

For inquiries regarding the trigger input signal (option), please contact SHIBAURA MACHINE Customer Service or the Headquarters Sales Department.

Controller connector: JEZ-9S-3 * J.S.T.



Pin No.	Signal name	Signal description	Pin No.	Signal name	Signal description
1	LI0	Trigger input 0	6	LI1	Trigger input 1
2	LI2	Trigger input 2	7	LI3	Trigger input 3
3	FUGND	P24G ground	8	COM	Trigger input common
4	EXP24V	External power supply P24V	9	NC	—
5	FG	FG			

11.1 Circuit Configuration

Since the input circuit uses a bidirectional optocoupler, it is possible to switch between Type N and Type P.

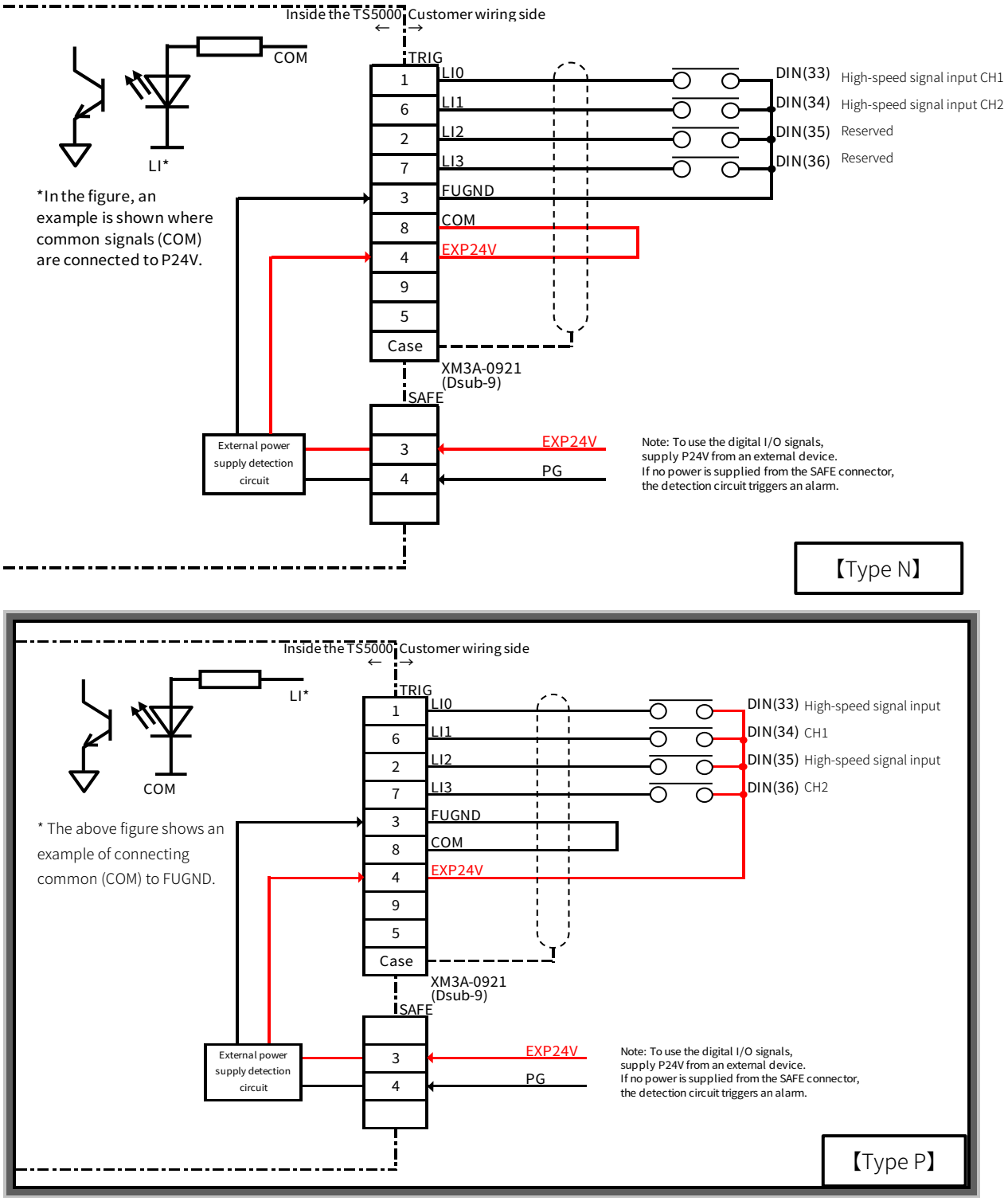


Fig. 11.1 Connecting the Trigger Input Signal Cable

11.2 Fabricating the TRIG Connector Cable

When fabricating a trigger input signal cable, observe the following matters.

a.) Use the optionally supplied connector or a connector of the same type for the TRIG connector of the controller. The equivalent of other makers (D-sub connectors) can also be used.

Connector name	Type of connector	Maker
TRIG	XM3A-0921 (plug type connector) XM2S-0911 (connector cover)	OMRON

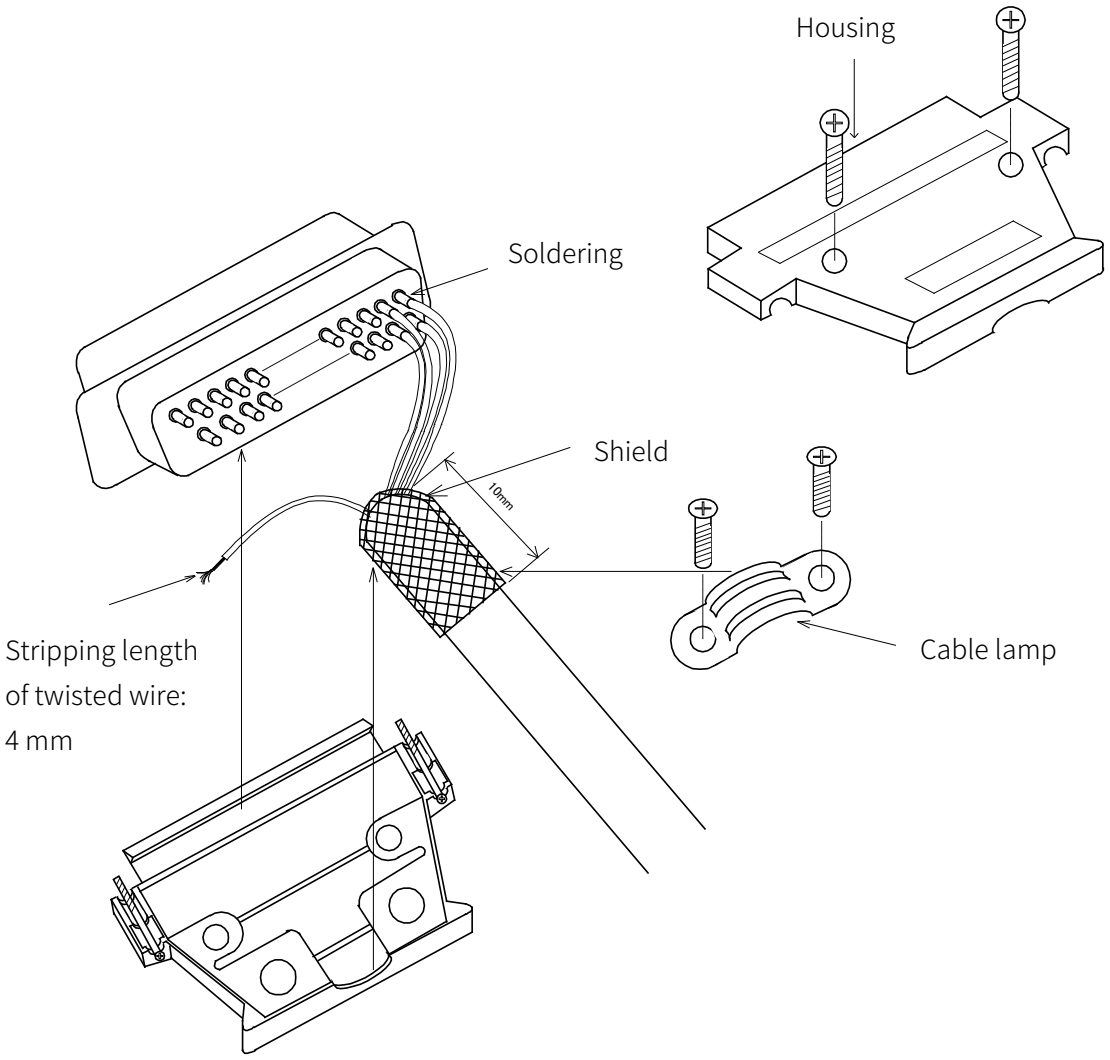
b.) Select the cables which meet the following specifications.

- Core wire : 0.18 mm^2 to 0.32 mm^2 twisted wire, Withstand voltage 60V/1A or less/75 ℃
- Outer diameter of cable used : Maximum φ10 mm
- Shield : Batch braided shield
- Cable length : 30m or less

⚠DANGER

- Never use unauthorized cables. If a cable that does not meet the specifications is used, a fire can occur due to a short circuit or overheating of the core wire. In addition, the robot may malfunction due to noise entry, resulting collision with the robot.

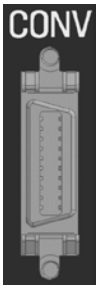
c.) The TRIG connector is a solder cup type where the wire is joined by soldering.
Fabricate the connection cables for the input/output signal connectors OUTPUT and OUTPUT as shown in the drawing below.
The shield of the cable using the D-sub connector is connected to the ground of the robot controller by passing through the housing. Completely connect the shield wire to the housing using the cable clamp. Also, it is recommended to attach a protective cover such as thermal contraction tube to the soldered part to prevent a short-circuit.



12. CONV Slot (Option)

The CONV slot is for extension of the conveyor synchronization encoder input. A signal cable for the conveyor synchronization function (supporting 2 channels of incremental encoder input for conveyor synchronization) can be added as an option. Use an encoder with A and B phase signals and a phase difference of 90 degrees, and with an RS-422 differential output (line driver) output circuit. In addition, the maximum response frequency of the A/B phase is 250 kHz with wire breakage detection.
Use the optional connector [PCR-E20FS+ (connector), PCR-V20LB (hood)] to connect the signal cables for the conveyor synchronization function.
For details on vision + conveyor synchronization, refer to “STEA1183_Vision Sensor + Conveyor Synchronization Function Manual”.
For inquiries regarding the conveyor synchronization function signal (option), please contact SHIBAURA MACHINE Customer Service or the Headquarters Sales Department.

Controller connector: PCR-E20PMRS+ * Honda Tsushin Kogyo Co., Ltd.



Pin No.	Signal name	Signal description	Pin No.	Signal name	Signal description
1	CH1AH	Conveyor A phase H	11	CH2AH	Conveyor A phase H
2	CH1AL	Conveyor A phase L	12	CH2AL	Conveyor A phase L
3	CH1BH	Conveyor B phase H	13	CH2BH	Conveyor B phase H
4	CH1BL	Conveyor B phase L	14	CH2BL	Conveyor B phase L
5	P5V	P5V power supply	15	P5V	P5V power supply
6	GND	P5G ground	16	GND	P5G ground
7	NC	—	17	NC	—
8	NC	—	18	NC	—
9	NC	—	19	NC	—
10	NC	—	20	NC	—

12.1 Connecting Incremental Encoders to Device

Use shielded twisted pair cables as shown below. Also, connect in pairs as shown in the figure. Please note that if you make a mistake in the connection procedure, the noise resistance will deteriorate and it may cause a malfunction. Also, refer to Fig. 12.1 for the connection when using the external power supply of the robot controller, and Fig. 12.2 when using the internal power supply. (Since there are two channels of encoder input, use them as needed.)

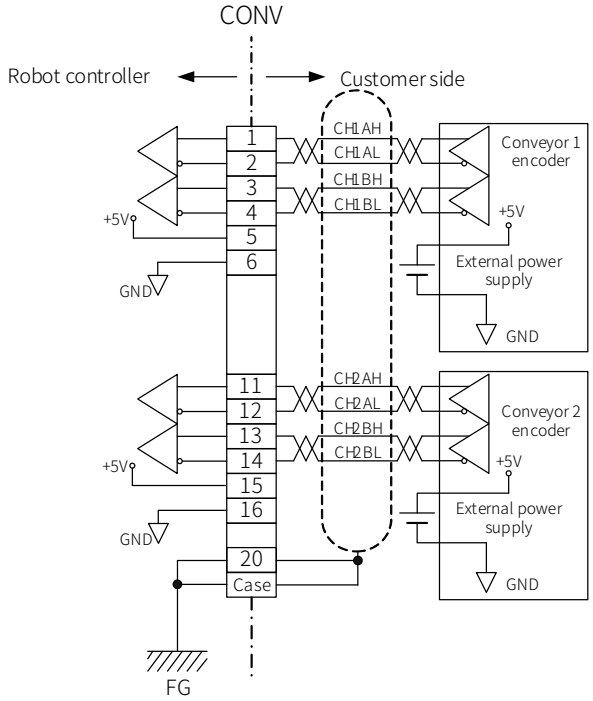


Fig. 12.1 Connecting Incremental Encoders to Device (Using External Power Supply): Recommended

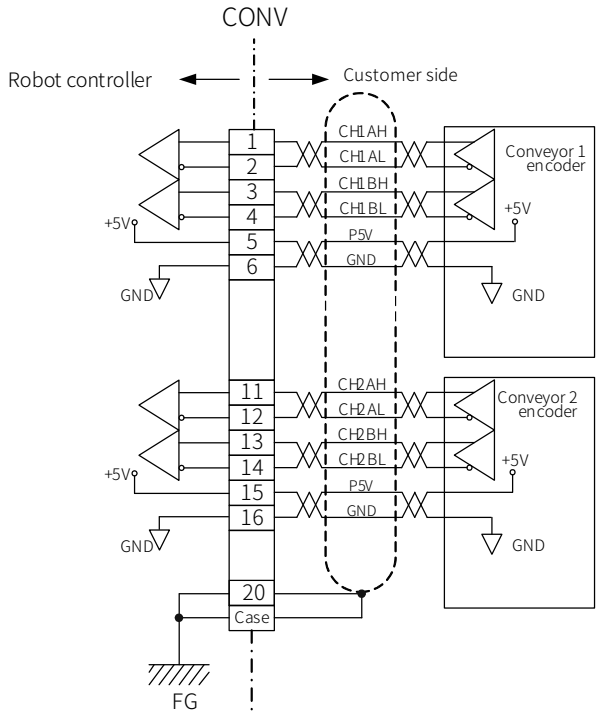


Fig. 12.2 Connecting Incremental Encoders to Device (Using Internal Power Supply): Not Recommended

12.2 Fabricating the CONV Connector

When fabricating conveyor encoder cables, observe the following matters.

a.) Use the optionally supplied connector or a connector of the same type for the CONV connector of the controller.

Connector name	Type of connector	Maker
CONV	PCR-S20FS+ (connector) PCR-V20LB (connector cover)	Honda Tsushin Kogyo Co., Ltd.

b.) Select the cables which meet the following specifications.

- Core wire used : 0.2 mm^2 to 0.52 mm^2 twisted wire, Withstand voltage 60 V/1 A or less/75 ℃
- Outer diameter of cable used : Maximum φ8.0 mm
- Shield : Batch braided shield
- Cable length : Refer to 12.2.1 Relationship between encoder cable length and number of power supply twisted pairs.

⚠DANGER

- Never use unauthorized cables. If a cable that does not meet the specifications is used, a fire can occur due to a short circuit or overheating of the core wire. In addition, the robot may malfunction due to noise entry, resulting collision with the robot.

c) How to fabricate a cable

The connector is a solder cup type that joins electric wires by soldering. Pay attention to the method of joining the electric wire to the connector and the method of shielding, and fabricate connectors by referring to Fig. 12.3.

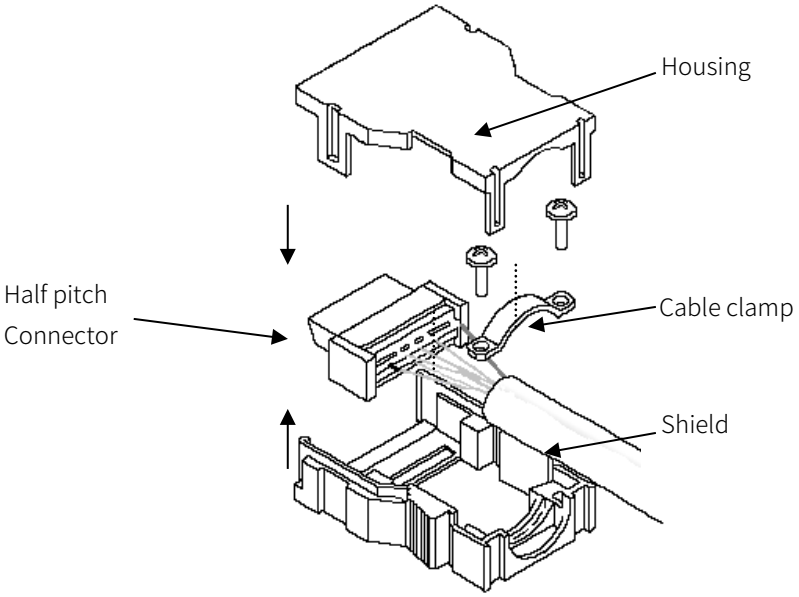


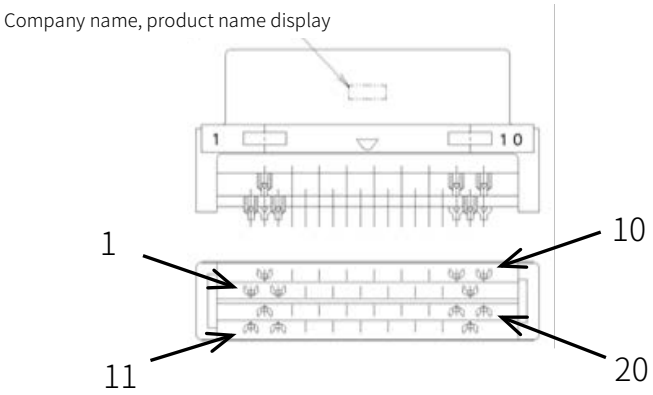
Fig. 12.3 How to Fabricate a Cable

MEMO

・ The shield of the cable is connected to the ground of the robot controller by passing through the housing. Completely connect the shield wire to the housing using the cable clamp.

c.) The CONV connector is a type that joins electric wires by soldering.

d.) Connector pin arrangement



12.2.1 Relationship between encoder cable length and number of power supply twisted pairs

When using +5 V supplied from the robot controller as the power supply for the encoder, connect the wires in consideration of the voltage drop in the cable.

<Example of cable voltage drop calculation>
When the allowable voltage of the encoder is 4.5 V, the relationship between the cable length and the number of twisted pairs of power supply is as follows. (Calculate the wire resistance as 110 Ω/km, encoder current consumption 150 mA, controller supply voltage 5 V.)

$$V_e = 5 - \frac{110}{1000} \times l \times 2 \times \frac{150}{1000} \times \frac{1}{p}$$

Here, V_e: encoder voltage, l: cable length, 2: number of round trips, p: number of twisted pairs are shown.
From the above formula, the cable length should be less than or equal to the following length in order to secure the allowable voltage 4.5 V for the encoder.

- 15 m or less when the number of power supply twisted pairs is 1.
- 30 m or less when the number of power supply twisted pairs is 2.
- 45 m or less when the number of power supply twisted pairs is 3.

Therefore, if you need a cable of 45 m or more, use an external power supply.

However, the above calculation example is limited to the above conditions. Please calculate by yourself referring to the above calculation example. In particular, please note that the current consumption varies depending on the encoder used. It is possible to increase the cable length by using an external power supply, but 100 m is the maximum length due to the signal.

I/O Cables

This describes the I/O cables connected to the controller.

13. I/O Cables

Controller maintenance and inspection must be performed by qualified personnel.
Also, these processes must be compliant with the laws, regulations, and ordinances of the respective country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

13.1 INPUT Cable

13.1.1 Specifications

Type : 368M54____
↑
Symbol in the table below.

* To designate the cable length (L), see the table below.

Symbol	L (m)	Symbol	L (m)
01	3	06	12
02	5	07	15
03	6		
04	8		
05	10		

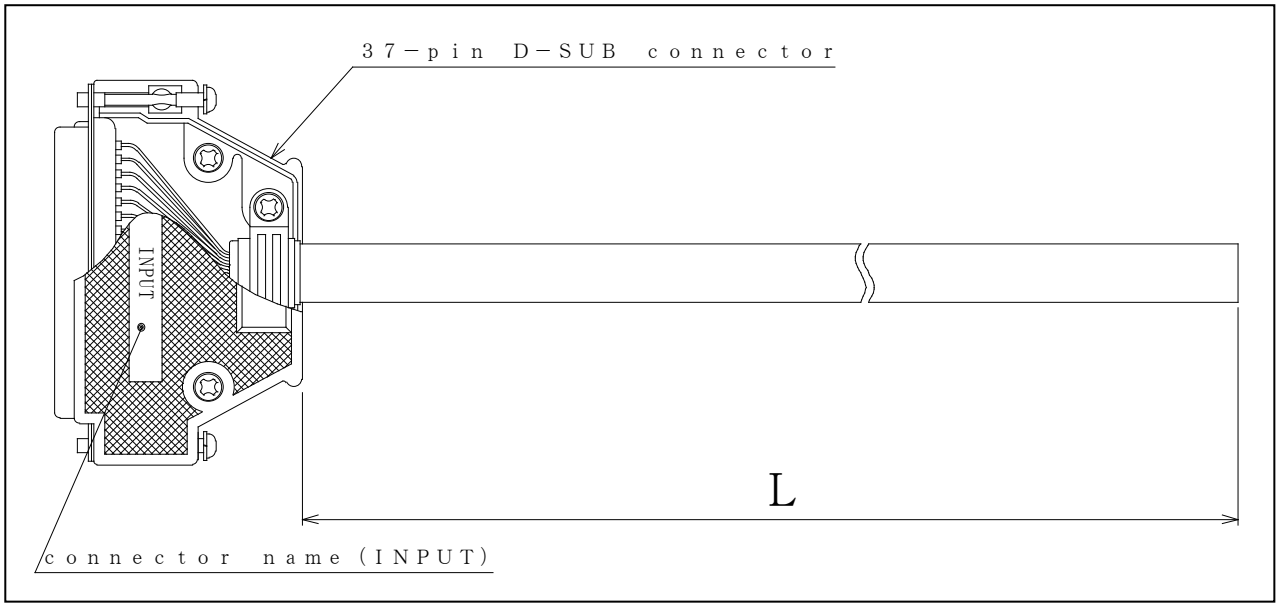
* For details on the input signals and the connection method, refer to 5. Input Signal Cable.
* This cable cannot be used for the sliding part or moving part which will bend repeatedly.
* For details, please contact SHIBAURA MACHINE Customer Service or Sales Department at the headquarters.

Cable specifications

Items	Specifications
Nominal sectional area of conductor	0.2 mm^2
Structure of conductor	7/0.18
No. of cable cores	40C
Insulator material	Heat-proof vinyl chloride
Shield material	Tin-plated annealed copper wire
Sheath material	Heat-proof vinyl chloride
Conductor resistance (20°C)	119 Ω/km or less
Operating temperature	–20°C to 75°C[1)]
Operating voltage	50 VAC, 60 VDC

1) Heat-proof grade: 90°C

13.1.2 External View



13.1.3 Cable Connection Table

Signal name	Extension Input signal name	Connector pin	Cable color
DI_1	DI_9	1	Black
DI_2	DI_10	20	White
DI_3	DI_11	2	Red
DI_4	DI_12	21	Green
DI_5	DI_13	3	Yellow
DI_6	DI_14	22	Brown
DI_7	DI_15	4	Blue
DI_8	DI_16	23	Gray
P24V	P24V	5	Orange
P24G	P24G	24	Purple
INCOM	INCOM1	6	Pink
STROBE	DI_17	25	Light green
PRG_RST	DI_18	7	Light blue
STEP_RST	DI_19	26	Black/White
CYC_RST	DI_20	8	Red/White
DO_RST	DI_21	27	Green/White
ALM_RST	DI_22	9	Yellow/White
RUN	DI_23	28	Brown/White
EX_SVON	DI_24	10	Blue/White
P24V	P24V	29	Gray/White
P24G	P24G	11	Orange/White
SYSINCOM	INCOM2	30	Purple/White
STOP	DI_25	12	Pink/White
CYCLE	DI_26	31	Light green/White
LOW_SPD	DI_27	13	Light blue/White
BREAK	DI_28	32	White/Black
SVOFF	DI_29	14	Red/Black
-	-	33	Green/Black
-	-	15	Yellow/Black
-	-	34	Brown/Black
P24V	P24V	16	Blue/Black
P24G	P24G	35	Gray/Black
SYSINCOM	INCOM2	17	Orange/Black
-	-	36	Purple/Black
-	-	18	Pink/Black
-	-	37	Light green/Black
-	-	19	Light blue/Black
FG	FG	Case	Shield

Type N

Type P

13.2 OUTPUT Cable

13.2.1 Specifications

Type : 368M55___
↑
Symbol in the table below.

* To designate the cable length (L), see the table below.

Symbol	L (m)	Symbol	L (m)
01	3	06	12
02	5	07	15
03	6		
04	8		
05	10		

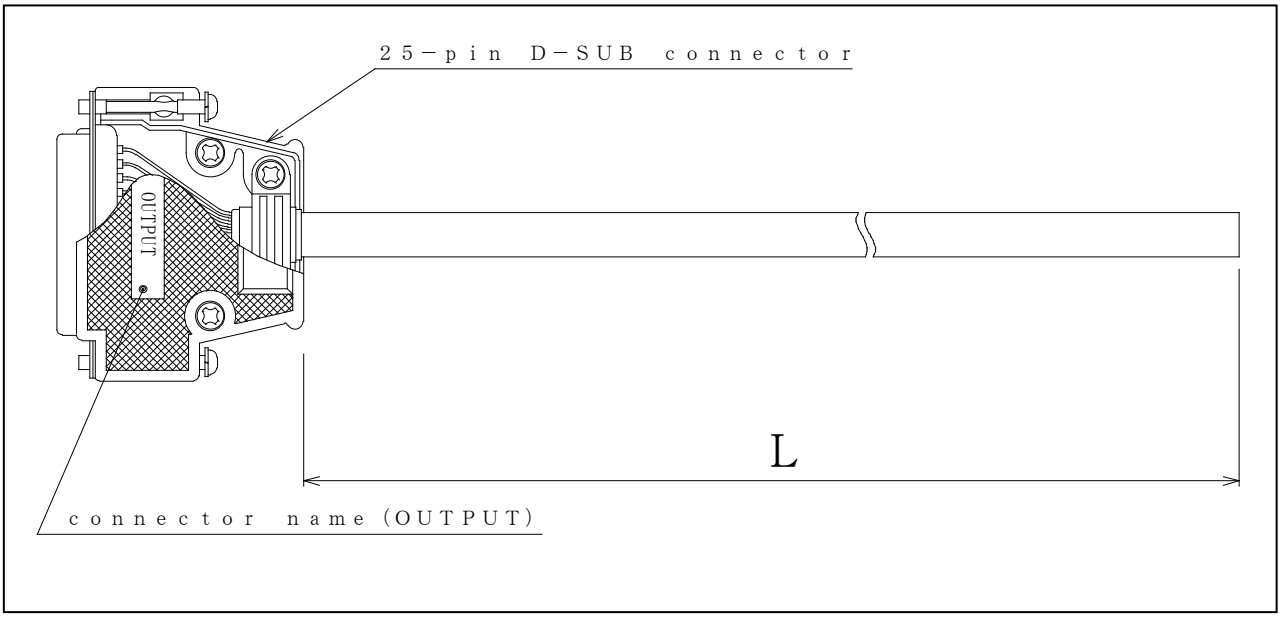
* For details on output signals and the connection method, refer to 6 Output Signal Cable.
* This cable cannot be used for the sliding part or moving part which will bend repeatedly.
* For details, please contact SHIBAURA MACHINE Customer Service or Sales Department at the headquarters.

Cable specifications

Items	Specifications
Nominal sectional area of conductor	0.2 mm^2
Structure of conductor	7/0.18
No. of cable cores	25C
Insulator material	Heat-proof vinyl chloride
Shield material	Tin-plated annealed copper wire
Sheath material	Heat-proof vinyl chloride
Conductor resistance (20°C)	119 Ω/km or less
Operating temperature	–20°C to 75°C [1)]
Operating voltage	50 VAC, 60 VDC

1) Heat-proof grade: 90°C

13.2.2 External View



13.2.3 Cable Connection Table

Signal name	Connector pin	Cable color	Signal name
DO_1	DO_9	1	Black
DO_2	DO_10	14	White
DO_3	DO_11	2	Red
DO_4	DO_12	15	Green
DO_5	DO_13	3	Yellow
DO_6	DO_14	16	Brown
DO_7	DO_15	4	Blue
DO_8	DO_16	17	Gray
OUTCOM	OUTCOM1	5	Orange
OUTCOM	OUTCOM1	18	Purple
ACK	DO_17	6	Pink
SV_RDY	DO_18	19	Light green
EXTSIG	DO_19	7	Light blue
SYS_RDY	DO_20	20	Black/White
AUTORUN	DO_21	8	Red/White
CYC_END	DO_22	21	Green/White
LOW_ST	DO_23	9	Yellow/White
BT_ALM	DO_24	22	Brown/White
SYSOUTCOM	OUTCOM2	10	Blue/White
SYSOUTCOM	OUTCOM2	23	Gray/White
ALARM	DO_25	11	Orange/White
P24V	P24V	24	Purple/White
P24V	P24V	12	Pink/White
P24G	P24G	25	Light green/White
P24G	P24G	13	Light blue/White
FG	FG	Case	Shield

Type N

Type P